SECJ2203: Software Engineering

System Documentation (SD)

QMS-Flow

Version 1.0

10/12/2025

Faculty of Computing

Prepared by: QualiTech Solutions

Revision Page

a. **Overview**

This version (1.0) of the System Documentation (SD) focuses on the System Requirements Specification (SRS) component, detailing the introduction and specific requirements for the QMS-Flow system. It includes personas, user stories, system features with modular structure, launch phases, performance requirements, and design constraints. This aligns with the Agile Scrum process model selected in the project proposal, emphasizing iterative development for ISO 9001:2015 compliance.

b. **Target Audience**

This SD is intended for the following key audiences, each with specific roles in the development, use, or evaluation of the QMS-Flow system:

- Development Team (QualiTech Solutions): Includes team members responsible for analyzing requirements, designing, and implementing the system to ensure alignment with specified features.
- Admins at Company XYZ Sdn. Bhd.: System administrators who will use the document to configure settings and manage user roles for secure operations.
- Quality Managers at Company XYZ Sdn. Bhd.: End-users who will review the document for compliance with ISO 9001:2015 processes and provide feedback on usability in daily quality oversight.
- Department Heads at Company XYZ Sdn. Bhd.: Supervisory users who approve changes and manage findings, referencing the SD for workflow enforcement.
- Employees at Company XYZ Sdn. Bhd.: Operational users who interact with modules like complaints logging, benefiting from clear requirements to ensure the system supports efficient workflows.
- Top Management at Company XYZ Sdn. Bhd.: Stakeholders focused on strategic oversight, using the SD to verify how the system supports business objectives like risk reduction and continual improvement.
- External Auditors: Independent reviewers verifying ISO 9001:2015 compliance, who will reference the document to assess traceability and adherence to standards.

c. **Project Team Members**

Member Name	**Role**	**Task**	**Status**
LEE HUI ZHEN (A24CS0259)	Team Leader	Sec 2.2: System Features Module 4: Corrective Actions (CAPA)	Completed
MUHAMMAD LUQMAN BIN MOHD AZMI (A24CS0138)	System Analyst	Sec 1.1 & 1.2: Purpose & Scope Module 1: Audit Management Sec 2.5: Performance Requirements	Completed
SEAH ZHANG JIAN (A24CS0297)	System Architect	Sec 1.3 & 1.5: Definitions & Overview	Completed

		Module 2: Change Control Sec 2.6: Design Constraints	
TAN HUI SHAN (A24CS0197)	Business Analyst	Sec 2.1: Personas Sec 2.3: Launch Phase Module 3: Customer Complaints	Completed

d. **Version Control History**

Version	**Primary Author(s)**	**Description of Version**	**Date Completed**
1.0	Team leader 1 (LEE HUI ZHEN)	Completed Chapter 1 (Introduction) and Chapter 2 (Specific Requirements)	10/12/2025
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1. Introduction

1.1 Purpose

This System Documentation outlines the complete specification, design, and testing approach for the QMS-Flow system, which is a web based Quality Management System developed for the Company XYZ Sdn. Bhd. to achieve and maintain ISO 9001:2015 compliance.

This System Documentation contains three main components:

- **System Requirements Specification (SRS)** - detailing functional and non-functional requirements.
- **System Design Documents (SDD)** - describing architectural and component design.
- **System Testing (STD)** - outlining validation and verification strategies.

This documentation is intended for development teams, system administrators, quality managers, department heads, employees, top management, and external auditors to ensure a shared understanding of system capabilities, constraints, and compliance alignment.

1.2 Scope

This system documentation refers to the QMS-Flow software system, a centralized web platform for managing ISO 9001:2015 compliance processes at Company XYZ Sdn. Bhd. It replaces manual, paper-based workflows with automated modules for audit management, document control, complaint handling, and corrective actions (CAPA).

Some of the In-scope functionalities are :

- Schedule and track internal audits.
- Upload, version, and approve documents (SOPs) with automated workflows.
- Log and investigate customer complaints.
- Initiate, assign and verify CAPA records.
- Generate compliance reports and maintain audit trails.

Out-of-scope items:

- Integration with financial or ERP systems.
- Native mobile applications (system is only web-responsive)
- Real-time communication or chat features.

In short, the system will be developed using an Agile Scrum methodology, ensuring iterative delivery and stakeholder, which will ensure iterative delivery and stakeholder feedback. It will be compatible with modern web browsers and integrate with the existing LDAP/ Active Directory for authentication.

1.3 Definitions, Acronyms and Abbreviation

Term / Acronym	**Definition**
QMS	Quality Management System. A formalized system that documents processes, procedures, and responsibilities for achieving quality policies and objectives.
QMS-Flow	The web-based software system designed to manage ISO 9001:2015 compliance for Company XYZ Sdn. Bhd.
ISO 9001:2015	The international standard that specifies requirements for a quality management system (QMS).
CAPA	Corrective and Preventive Action. A process used to identify, investigate, and correct problems to prevent their recurrence.
SOP	Standard Operating Procedure. A set of step-by-step instructions compiled by an organization to help workers carry out complex routine operations.
RCA	Root Cause Analysis. A method of problem-solving used for identifying the root causes of faults or problems.
Admin	System Administrator responsible for managing user accounts and system configuration.
QM	Quality Manager. The primary user responsible for overseeing audits, document control, and CAPA workflows.
SD	System Documentation. The comprehensive document contains requirements, design, and testing details.
SRS	System Requirements Specification. A document that describes the nature of a project, software, or application.

SDD	System Design Document. A document that describes how the software will be structured to satisfy the requirements.
STD	System Testing Document. A document that outlines the testing strategy, cases, and results for the system.
LDAP	Lightweight Directory Access Protocol. Used for integrating user authentication with Company XYZ's existing directory services.

1.4 References

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1.5 Overview

This **System Documentation (SD)** focuses specifically on the **System Requirements Specification (SRS)** for the **QMS-Flow** system. Building upon the foundation established in the Project Proposal (Assignment 1), this document provides detailed, actionable requirements for developing a web-based Quality Management System that enables Company XYZ Sdn. Bhd. to achieve and maintain ISO 9001:2015 compliance.

The SRS is organized to comprehensively capture both functional and non-functional requirements:

- **Section 1 (Introduction)** establishes the document's purpose, scope, definitions, and references, providing contexts for all stakeholders including developers, quality managers, and external auditors.
- **Section 2 (Specific Requirements)** forms the core of this document, detailing:
    - **User Personas** representing key stakeholders (Quality Manager, Auditor, Department Head, Employee)
    - **System Features** described through Use Case Diagrams and detailed module descriptions
    - **User Stories** with acceptance criteria, activity diagrams, and sequence diagrams
    - **Performance Requirements** defining system responsiveness, capacity, and reliability
    - **Design Constraints** outlining technical, security, and compliance limitations

This SRS serves as the primary reference for the development team (QualiTech Solutions) during the Agile Scrum sprints outlined in the Project Proposal. It ensures all requirements are traceable to the identified business needs, particularly the automation of audit management, document control, CAPA processes, and compliance tracking while providing measurable criteria for system validation and acceptance testing.

2. Specific Requirements

The Specific Requirements section defines all functional and non-functional requirements of the QMS-Flow system. It outlines how the system should behave from the user's perspective, including the inputs, outputs, and operations needed to support ISO 9001:2015 compliance. This section also identifies the intended users through detailed personas and presents user stories that capture their needs and expectations. These requirements provide a clear foundation for designing, developing, and validating the QMS-Flow system.

2.1 Persona

The Personas section describes the key user groups who will interact with QMS-Flow. Each persona represents a typical user with specific goals, motivations, behaviours, and challenges related to quality management processes. Understanding these personas ensures that the system is designed to meet real user needs and supports effective adoption across the organization.

2.1.1 Persona 1 (Quality Manager)

2.1.1.1 Description

Amina Rahman, a 42-year-old Quality Manager at Company XYZ, oversees all ISO 9001:2015 compliance activities including internal audits, CAPA processes, document control, and management review documentation. She has solid experience with digital systems and prefers structured workflows that reduce administrative workload. Amina is highly motivated to maintain the company's certification status and avoid external audit non-conformities. Her main goals are to gain real-time visibility over QMS performance, ensure departments close findings on time, and maintain an organized central repository of documents. Behaviorally, she checks the system dashboard multiple times a day, assigns tasks through automated reminders, and follows up on overdue items weekly. Her major pain points include scattered audit evidence, inconsistent document versions, difficulty tracking CAPA progress, and the heavy reliance on manual reminders when using spreadsheets and shared drives.

2.1.1.2 User Stories

- As a Quality Manager, I want to schedule internal audits so that all departments follow the QMS audit cycle.
- As a Quality Manager, I want to assign audit findings to users so that corrective actions are properly tracked.
- As a Quality Manager, I want to control document changes through a structured approval workflow so that all updates to SOPs are properly reviewed and traceable.
- As a Quality Manager, I want to track the investigation progress of each complaint so that I can ensure issues are resolved within the expected timeframe.
- As a Quality Manager, I want to link each CAPA to its root cause so that corrective actions address the underlying problem effectively.

- As a Quality Manager, I want to document the results of effectiveness checks so that future audits can verify that the CAPA actions have worked.

2.1.2 Persona 2 (Internal Auditor)

2.1.2.1 Description

Raymond Lim is a 35-year-old Internal Auditor with extensive knowledge of ISO 9001:2015 and its compliance requirements. He performs audits across various departments and prefers using tablets during inspections to log findings immediately. Raymond is motivated by efficiency and accuracy, aiming to complete audits with minimal administrative overhead. His goals include generating standardized checklists, ensuring findings are recorded consistently, and improving traceability of evidence. His behaviours include using templates, uploading photos directly during audits, and performing real-time classification of findings. Raymond's main pain-points stem from manual audit processes, such as version confusion between checklist files, misplacement of findings, and difficulty consolidating audit data across departments.

2.1.2.2 User Stories

- As an Auditor, I want to generate audit checklists from templates so that I can prepare audits quickly.
- As an Auditor, I want to record audit findings directly into the system so that documentation is accurate and centralized.
- As an Auditor, I want to upload photos or files as evidence so that findings are properly supported.
- As an Auditor, I want to classify each finding so that the Quality Manager can take appropriate action.

2.1.3 Persona 3 (Department Head)

2.1.3.1 Description

Siti Norhidayah, a 39-year-old Production Department Head, manages a large team on the factory floor and is responsible for ensuring her department complies with SOPs, training requirements, and QMS policies. Her motivations are to maintain smooth operations and avoid disciplinary or compliance issues. Her goals include timely closure of assigned CAPAs, ensuring her staff follow the latest SOPs, and keeping training records updated. She often checks the system upon receiving notifications and tends to rely on clear dashboards for quick updates. Behaviourally, she responds promptly when alerted, uses the system mostly through desktop computers, and reviews SOPs before briefing her team. Her pain-points include unclear SOP versioning, difficulty manually tracking employee competency, and missing deadlines due to unstructured manual tracking systems.

2.1.3.2 User Stories

- As a Department Head, I want to view all the CAPAs assigned to me so that I can prioritize actions.
- As a Department Head, I want to access the latest SOP versions so that my team follows the correct procedures.
- As a Department Head, I want to update the status of audit findings so that the Quality Manager is informed of progress.
- As a Department Head, I want to check employee training status so that I can ensure competency compliance.
- As a Department Head, I want to review and approve document changes so that only accurate and department-validated SOP updates are released

2.1.4 Persona 4 (Factory Operator-General Employee)

2.1.4.1 Descriptions

Amirul Hakimi, a 28-year-old factory operator, interacts with the QMS system mainly when he needs to review SOPs or report operational issues. He prefers simple interfaces and minimal steps as he is not highly tech-savvy. Amirul is motivated to do his job safely and correctly, with the goal of always using the correct, most updated procedure. He typically uses the system only occasionally, behaves cautiously when handling digital forms, and relies on straightforward navigation. His pain-points include difficulty locating correct SOP versions, lack of confidence with complex digital forms, and confusion caused by multiple document formats stored manually by supervisors.

2.1.4.2 User Stories

- As an Employee, I want to easily access the latest SOP so that I can follow correct work procedures.
- As an Employee, I want to log a complaint or issue so that management can address it.
- As an Employee, I want to see my training history so that I know if refresher training is required.

2.2 System Features

Product Perspective
QMS-Flow is a web-based software system that operates on modern web browsers (Chrome, Edge, Firefox) and is accessible via desktop and mobile devices. The system provides a centralized platform for Company XYZ to manage their Quality Management System (QMS), replacing manual spreadsheets and paper trails to ensure compliance with the ISO 9001:2015 standard. The system serves as a "single source of truth" for all quality data, utilizing a secure, role-based architecture to manage audits, documents, complaints, and corrective actions. The system also implements a real-time notification engine to alert users of upcoming deadlines and pending approvals.

Product Module and Function

The system features are illustrated in Figure 2.1 below. The detailed description of each module and functions is tabulated in Table 2.1.

Use Case Diagram

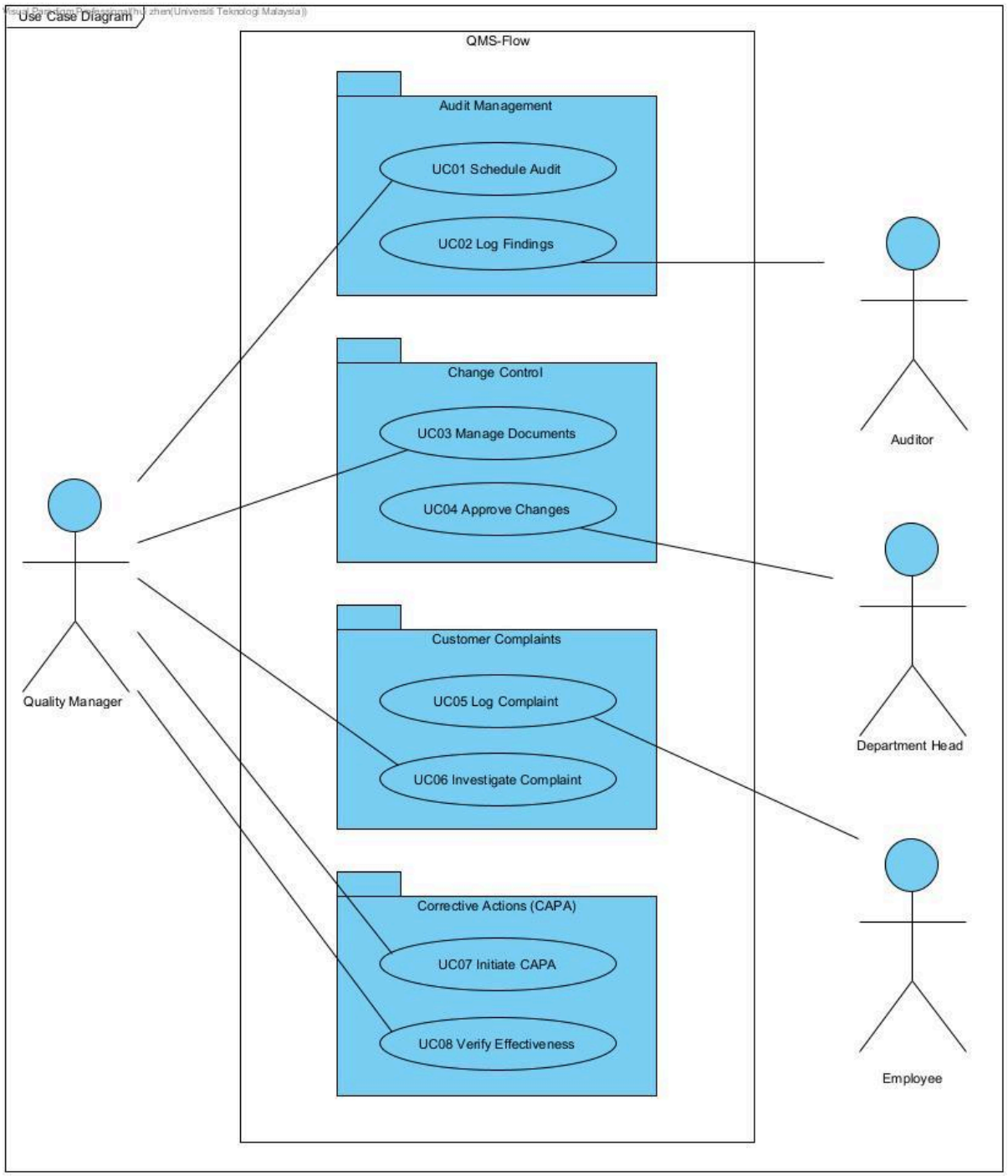

Figure 2.1: Use Case Diagram for QMS-Flow

Table 2.1: Description of Module and Functions for QMS-Flow

Use Case ID	Function	Description
UC01	Schedule Audit	This use case allows the Quality Manager to create an annual audit plan, scheduling audits for specific departments and assigning auditors
UC02	Log Findings	This use case allows Auditors to record audit observations or non-conformities directly into the system and assign them to department heads.
UC03	Manage Documents	This use case allows the Quality Manager to upload Standard Operating Procedures (SOPs) to a centralized repository with automated version control (e.g., v1.0, v1.1).
UC04	Approve Changes	This use case enables Department Heads to review and digitally approve document changes. During this process, the document is "locked" (read-only).
UC05	Log Complaint	This use case allows any Employee to log customer feedback or complaints, capturing details such as customer name, product, and nature of the issue.
UC06	Investigate Complaint	This use case allows the Quality Manager to assign a complaint to an investigator and set a target date for resolution.
UC07	Initiate CAPA	This use case allows the Quality Manager to create a Corrective Action (CAPA) record linked to a source (Audit Finding or Customer Complaint) to address root causes.
UC08	Verify Effectiveness	This use case allows the Quality Manager to record the results of effectiveness checks after a corrective action has been implemented.

Domain Model

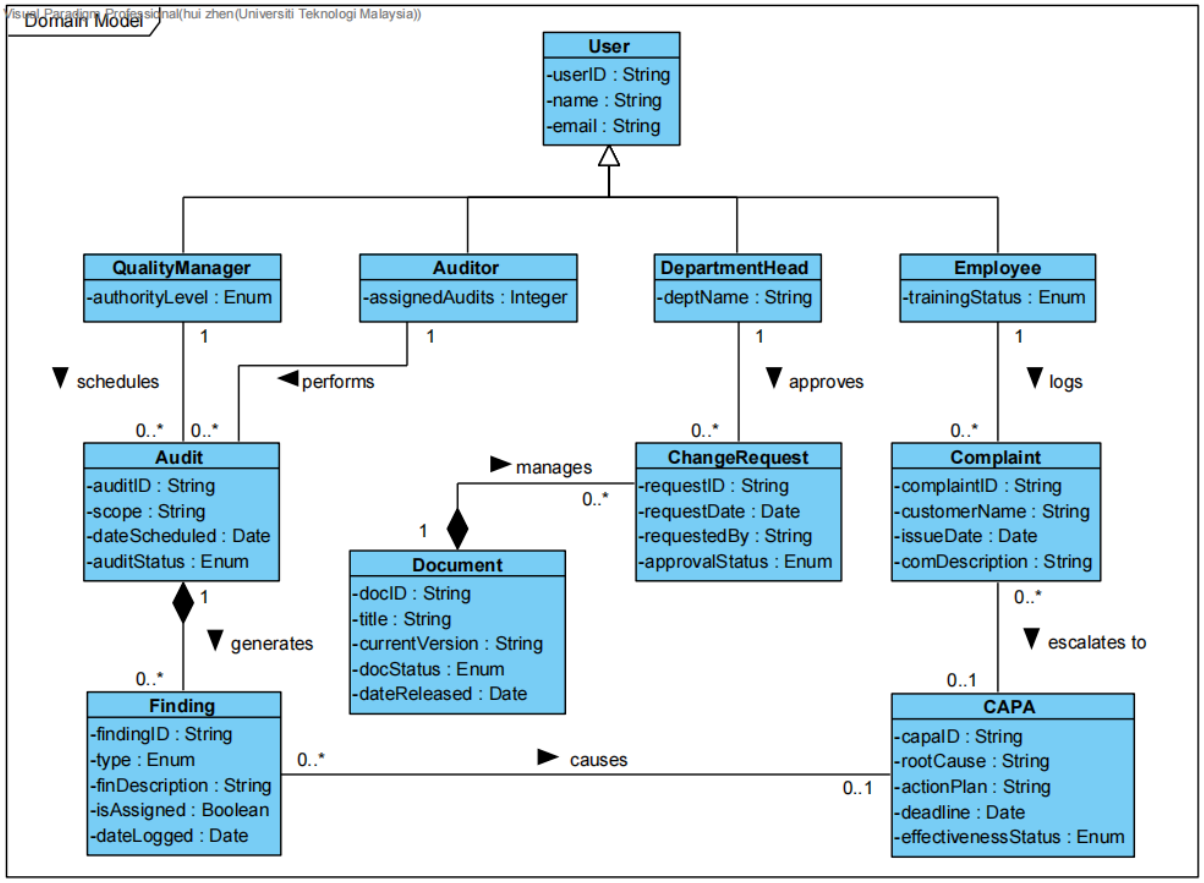

Figure 2.2: Domain Model for QMS-Flow

The Domain Model (Figure 2.2) defines the static structure and key business relationships between the core data entities of the QMS-Flow system. The design adheres to Object-Oriented principles and enforces strict quality management rules.

- Generalization (Inheritance): The design uses Generalization (inheritance) from the central User class to model the system's role-based access. This ensures that specialized actors (e.g., Auditor, QualityManager) inherit basic identity attributes while adding unique attributes (like certificationID or authorityLevel) required for their specific functions.
- Strong Ownership (Composition): Composition is utilized to enforce data integrity. For example, the Audit class strongly owns its associated Finding records, which means a Finding cannot exist unless it belongs to the specific Audit that generated it. This directly solves problems related to poor audit traceability. Similarly, Document strongly manages its history of ChangeRequest records.
- Process Flow (Association): Association models the quality workflow, where records from Complaint and Finding funnel into the central CAPA class. The 0..1 multiplicity on the CAPA side correctly indicates that the escalation to a formal corrective action is optional, depending on the severity of the issue.

State Machine Diagram

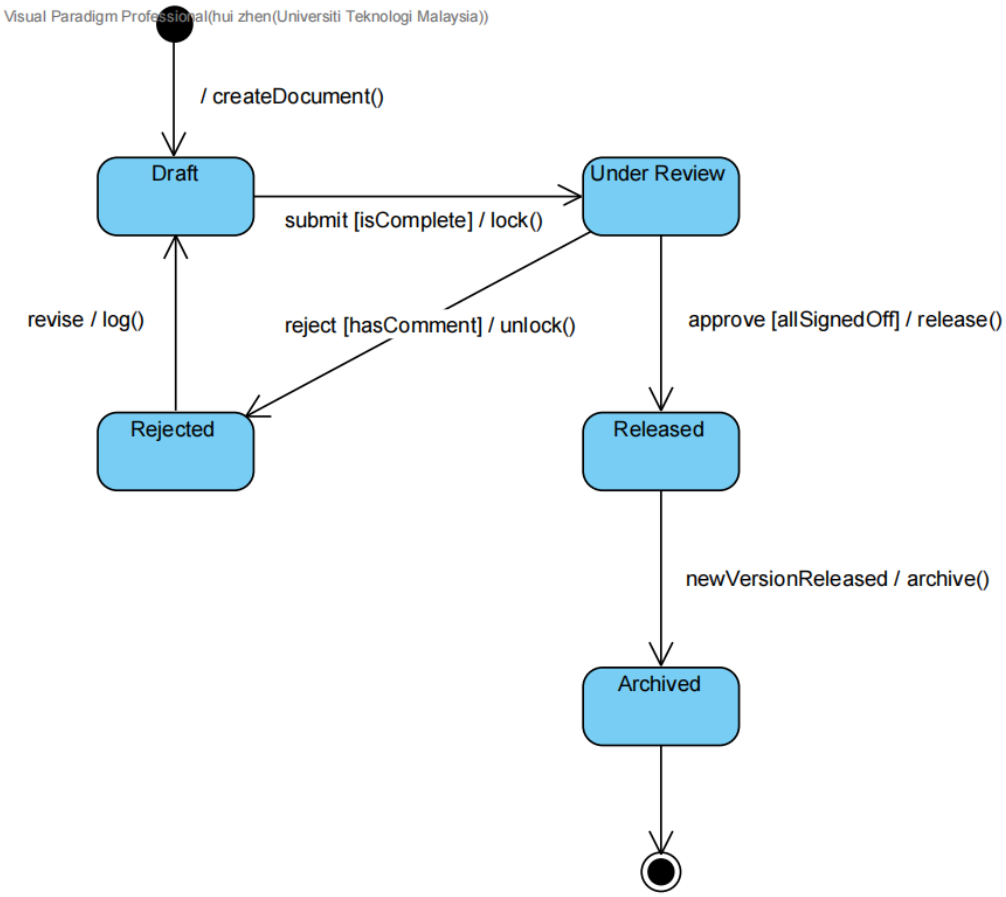

Figure 2.3: State Machine Diagram for QMS-Flow

The State Machine Diagram (Figure 2.3) models the specific, formal lifecycle of the Document class. This diagram is essential for enforcing the required Change Control workflow and maintaining version integrity.

- States and Control: The diagram clearly defines the five required states. The Under Review state is critical, as the document must be locked (read-only) while awaiting approval. The lifecycle concludes when the document enters the Archived state, typically after being superseded.
- Precision and Compliance: All transitions strictly follow the UML syntax Event [Guard Condition]/ Action to ensure unambiguous system behavior:
    - The submit event is conditional upon the [isComplete] guard before the system executes the lock() action.
    - The approve event is guarded by [allSignedOff] to prevent unauthorized or incomplete releases.
- Workflow Enforcement: The transitions enforce the mandatory revision loop: if the document is rejected, it must return to Draft for correction before it can be re-submitted for review.

2.3 Launch Phase

The launch phase establishes the initial product backlog for QMS-Flow, which contains all prioritized user stories and system features derived from the project's use cases and stakeholder needs. The backlog represents the core ISO 9001:2015 processes—such as audit management, document control, complaints handling, and CAPA—that must be implemented to deliver value to the Quality Manager, Auditor, Department Head, and Employee personas. Each backlog item is prioritized based on its importance, dependency, and contribution to compliance. This backlog then guides the sprint planning process, ensuring that high-priority and foundational features are developed first, followed by dependent modules in later sprints.

Sprint	**Team members assigned**	**User Stories Assigned**
Sprint #1	MUHAMMAD LUQMAN BIN MOHD AZMI	**Audit Management Module:** US001 - Schedule Audit US002 - Log Findings
Sprint #2	SEAH ZHANG JIAN	**Change Control Module:** US003 - Manage Documents US004 - Approve Changes
Sprint #3	TAN HUI SHAN	**Customer Complaints Module:** US005 - Log Complaint US006 - Investigate Complaint
Sprint #4	LEE HUI ZHEN	**Correctives Action (CAPA) Module:** US007 - Initiate CAPA US008 - Verify Effectiveness

2.4 User Story Details

2.4.1 US001: User Story <Schedule Audit>

Table 2.2: User Story Description for <Schedule Audit>

User story: <Schedule Audit>(UC01)
ID: US001
User Story Description As a Quality Manager, I want to create an annual audit plan, schedule audits for specific departments, and assign auditors so that the company can maintain a structured audit cycle and ensure ISO 9001:2015 compliance.
Pre-conditions: 1. The Quality Manager is authenticated and logged into the system. 2. The Quality Manager has the role permission "Quality Manager" or "Audit Manager". 3. The Audit Management module is accessible.
Flow of events: 1. The Quality Manager navigates to the Audit Management dashboard. 2. The system displays the current audit plan and an option to "Schedule New Audit". 3. The Quality Manager clicks "Schedule New Audit". 4. The system displays a form with fields: Audit Title, Department, Scheduled Date, Auditor, and Scope. 5. The Quality Manager fills in the required fields and selects an auditor from a dropdown list of available auditors. 6. The Quality Manager clicks "Save Audit". 7. The system validates the input (e.g., date not in the past, auditor is available). 8. The system saves the audit record and assigns it a unique Audit ID. 9. The system sends a notification to the assigned auditor. 10. The system logs the audit creation in the audit trail. 11. The system displays a success message: "Audit scheduled successfully." 12. Use case ends.
Alternative flow (A1):Auditor Unavailable If the selected auditor is already assigned to another audit on the same date (Step 7), the system displays an error: "Selected auditor is unavailable on the chosen date. Please select another auditor." The form remains open for correction. **Alternative flow (A1):Past Date Entered** If the scheduled date is in the past (Step 7), the system displays an error: "Audit date cannot be in the past. Please select a future date."
Postconditions: 1. A new audit record is created with status "Scheduled".

2. The assigned auditor receives a notification. 3. The audit is added to the annual audit plan and visible in the audit calendar. 4. The audit creation is logged in the system audit trail.
Acceptance Criteria 1. The system must prevent scheduling an audit with a past date. 2. The system must check auditor availability and prevent double-booking. 3. All mandatory fields (Audit Title, Department, Date, Auditor) must be validated before saving. 4. The audit must be assigned a unique Audit ID upon creation. 5. The assigned auditor must receive an automated notification (email/system alert).
Exception flow (E1): Database Connection Error If the system fails to save the audit due to a database error, the transaction is rolled back and an error message is displayed: "System error. Please try again." The form data is retained for re-submission. **Exception flow (E2): Form Timeout** If the session times out during form submission, the system redirects to the login page and all unsaved data is lost. Upon re-login, the user is returned to the Audit Dashboard.

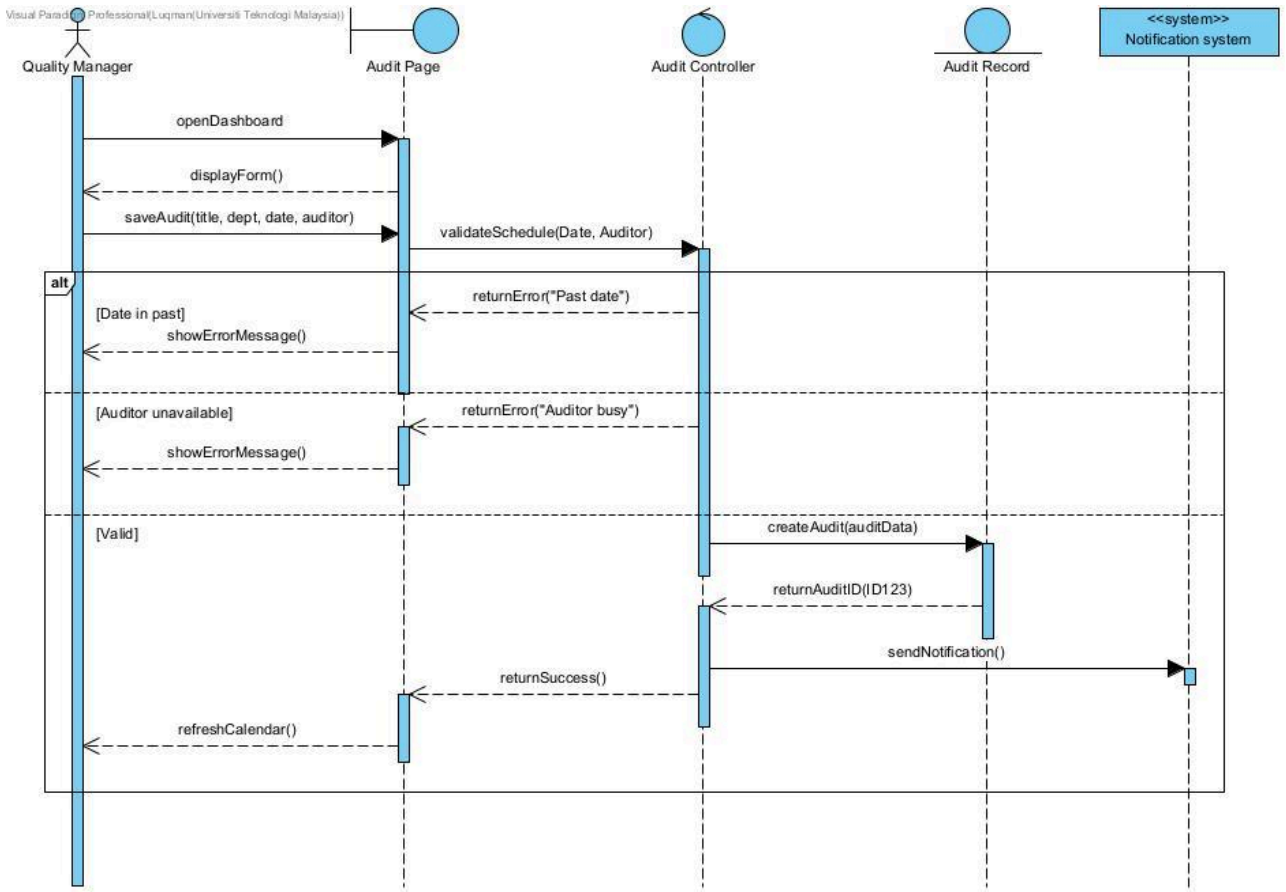

Figure 2.4: Sequence Diagram for <Schedule Audit>

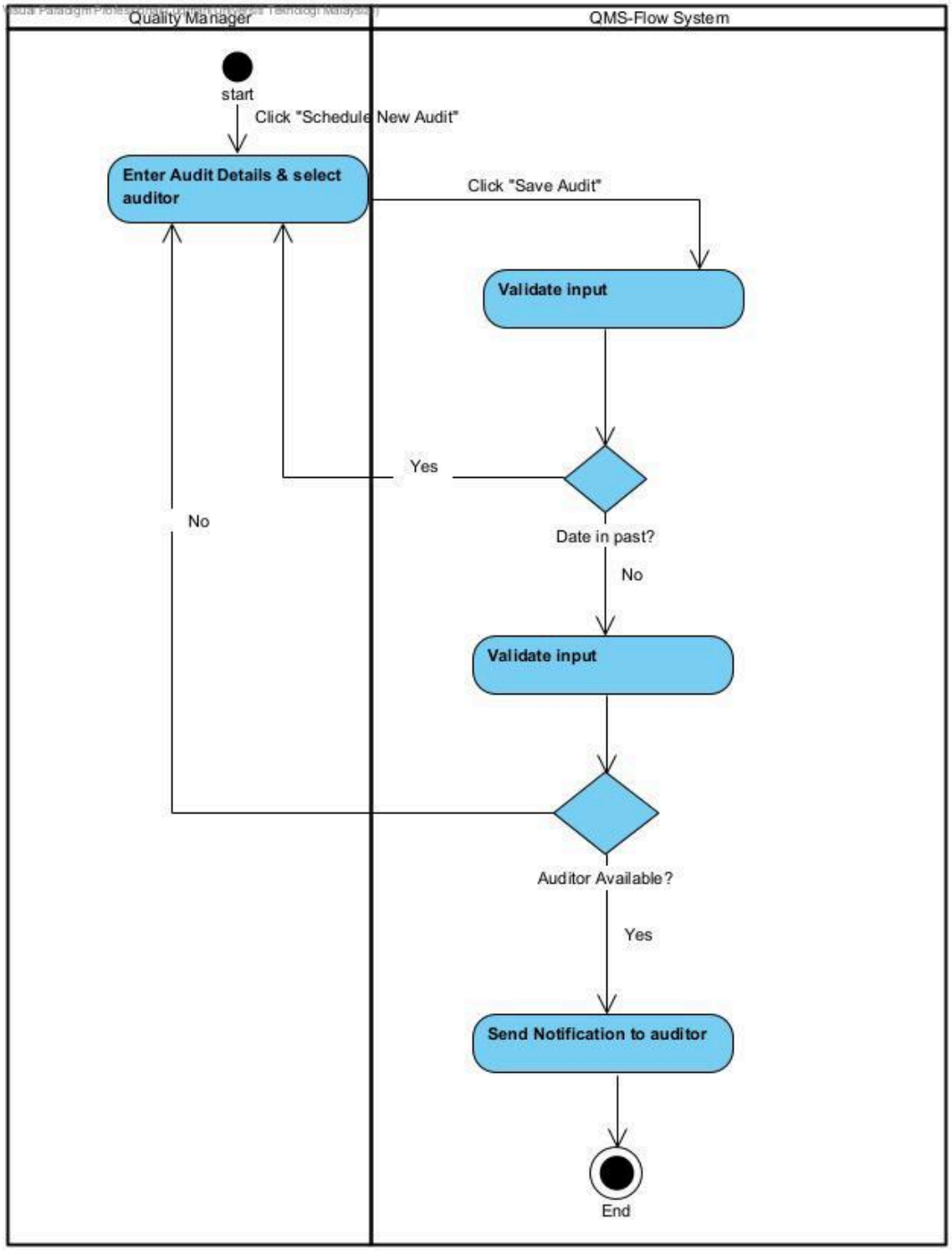

Figure 2.5: Activity Diagram for <<Schedule Audit>>

2.4.2 US002: User Story <Log Findings>

Table 2.3: User Story Description for <Log Findings>

User story: <Log Findings>(UC02)
ID: US002
User Story Description As an Auditor, I want to record audit findings (observations or non-conformities) directly into the system and assign them to department heads so that corrective actions can be formally tracked and resolved.
Pre-conditions: 1. The Auditor is authenticated and logged into the system. 2. The Auditor has been assigned to an audit that is in “In Progress” status. 3. The Audit Management module is accessible. 4. At least one audit is open and selectable.
Flow of events: 1. The Auditor navigates to the “My Audits” dashboard. 2. The system displays a list of assigned audits. 3. The Auditor selects an audit to log findings for. 4. The system opens the audit details page and displays an “Add Finding” button. 5. The Auditor clicks “Add Finding”. 6. The system displays a form with fields: Finding Description, Severity (Minor/Major/Critical), Department Affected, Evidence Upload (optional), and Assign To (dropdown of department heads). 7. The Auditor fills in the required fields and optionally attaches supporting files (photos, documents). 8. The Auditor clicks “Submit Finding”. 9. The system validates the input (e.g., description not empty, valid department head selected). 10. The system saves the finding and assigns it a unique Finding ID. 11. The system updates the audit’s findings list. 12. The system sends a notification to the assigned department head. 13. The system logs the finding creation in the audit trail. 14. The system displays a success message: “Finding logged successfully.” 15. Use case ends.
Alternative flow (A1): Missing Required Field If a required field is left empty (Step 9), the system highlights the missing field and displays an error: “Please fill in all required fields.” The form remains open. **Alternative flow (A2): File Too Large** If an uploaded file exceeds 50MB (Step 9), the system displays an error: “File size exceeds limit. Maximum size: 50MB.” The Auditor can remove or replace the file.
Postconditions: 1. A new finding record is created with status “Open”. 2. The assigned department head receives a notification.

3. The finding is linked to the audit and visible in the audit's findings log. 4. The finding creation is logged in the system audit trail.
Acceptance Criteria 1. Findings must be linked to a valid, active audit. 2. All mandatory fields (Description, Severity, Assign To) must be validated before submission. 3. File uploads must be limited to 50MB per file and accepted formats (PDF, JPG, PNG, DOCX). 4. Each finding must be assigned a unique Finding ID. 5. The assigned department head must receive an automated notification.
Exception flow (E1): Database Save Failure If the system fails to save the finding due to a database error, the transaction is rolled back and an error message is displayed: "Unable to save finding. Please try again." Form data is retained. **Exception flow (E2): Auditor Loses Access Mid-Process** If the Auditor's role permissions change or are revoked during the form filling, the system prevents submission and displays: "You no longer have permission to log findings." The user is redirected to the dashboard.

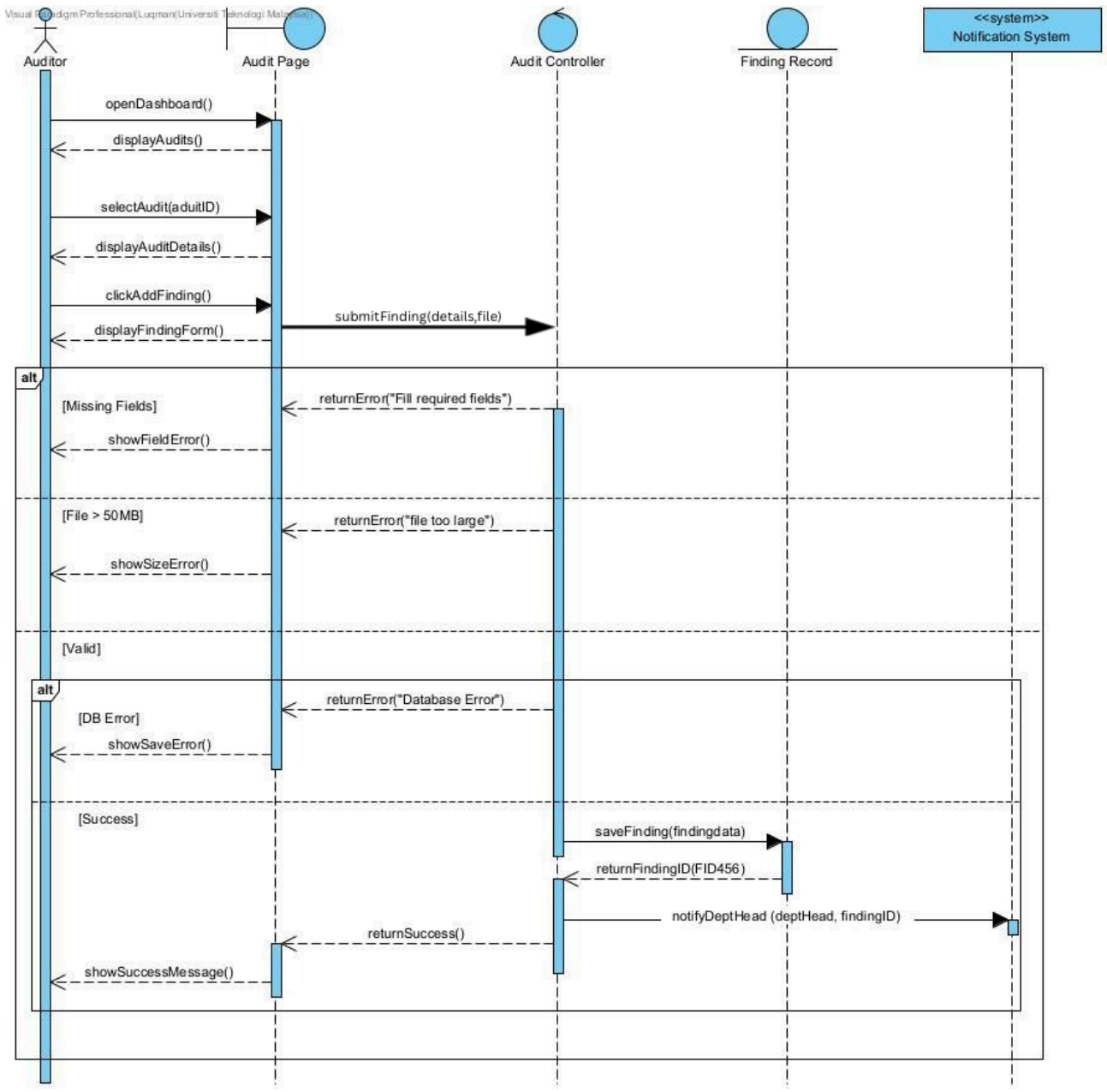

Figure 2.6: Sequence Diagram for <Log Findings>

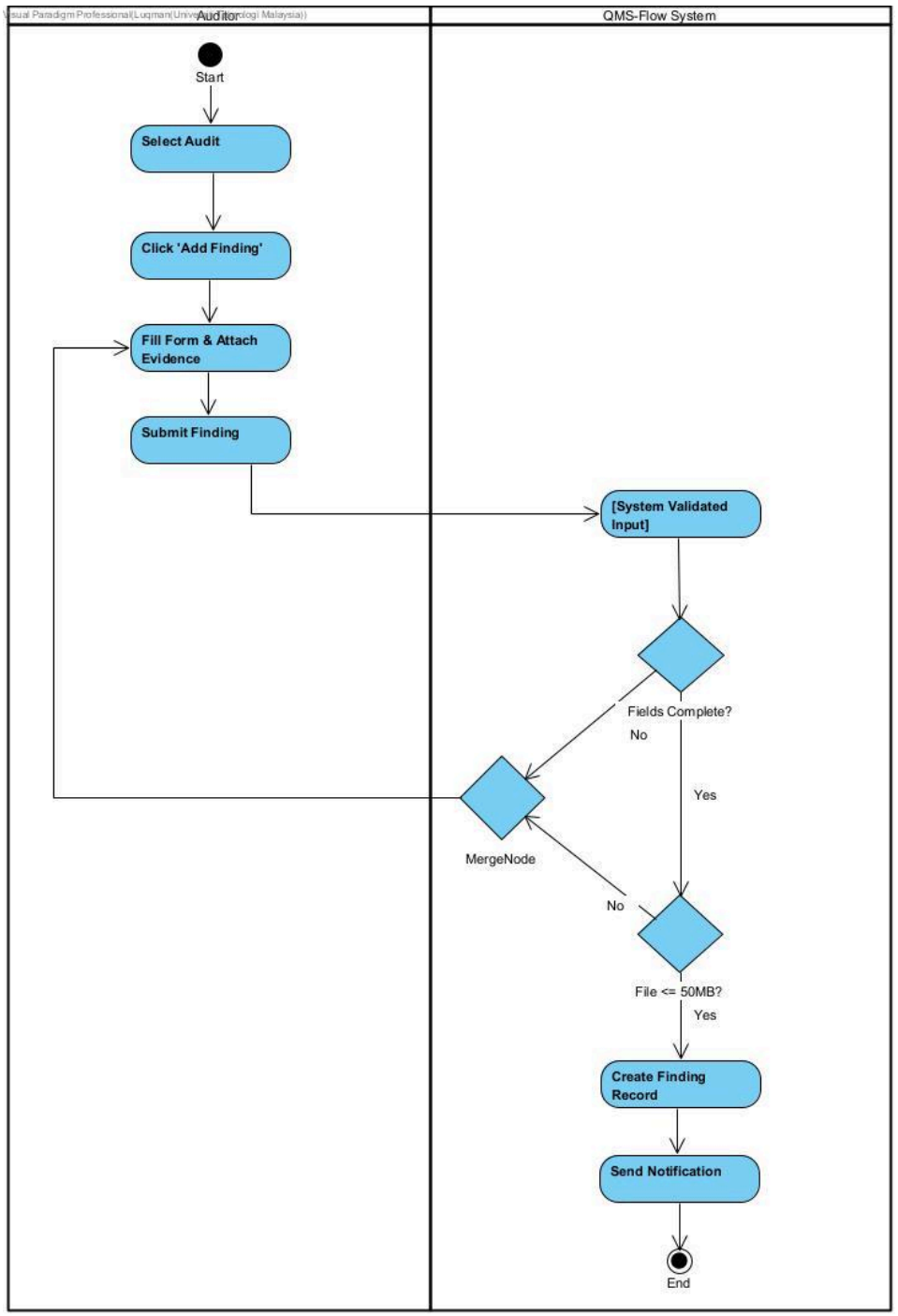

Figure 2.7: Activity Diagram for <Log Findings>

2.4.3 US003: User Story <Manage Documents>

Table 2.4: User Story Description for <Manage Documents>

User story: <Manage Documents> (UC03)
ID: US003
User Story Description As a Quality Manager, I want to upload Standard Operating Procedures (SOPs) to a centralized repository with automated version control so that all users access the latest approved documents and version history is maintained.
Preconditions: 1. Quality Manager is authenticated and logged in. 2. Users have "Document Controller" or "Quality Manager" role permissions. 3. The document repository module is accessible.
Flow of events: 1. Quality Manager navigates to the Document Control module. 2. The system displays the document repository with existing documents and an "Upload New Document" button. 3. Quality Manager clicks "Upload New Document". 4. The system displays an upload form with fields: Document Title, Document Type (SOP, Policy, Work Instruction), Department, Effective Date. 5. Quality Manager selects a file from local storage (PDF, DOCX, XLSX). 6. Quality Manager fills in required metadata and clicks "Submit". 7. System validates the file format (accepts .pdf, .docx, .xlsx, max 50MB). 8. System checks for duplicate document titles. 9. The system generates a new version number (e.g., v1.0) and saves the document. 10. The system logs the upload activity in the audit trail. 11. The system displays a success message. 12. Use case ends.
Alternative flow (A1): Duplicate Document Title If a document with the same title already exists (Step 8), system prompts: "Document '[Title]' exists. Do you want to create a new version?" If Yes, the system creates the next version (e.g., v1.1); if No, returns to the upload form. **Alternative flow (A2): Invalid File Format** If the uploaded file is not in accepted format (Step 7), the system displays error: "Invalid file format. Accepted formats: PDF, DOCX, XLSX."
Postconditions: **1**. New document is saved in a repository with status "Draft". 2. The document is automatically assigned the appropriate version number. 3. Audit trail records the upload with timestamp and user. 4. Document is locked for editing while in the repository.
Acceptance Criteria 1. System must accept only .pdf, .docx, and .xlsx file formats. 2. New documents must be assigned version v1.0; updated documents must increment version (v1.0 → v1.1). 3. All required metadata fields must be validated before submission. 4. System must detect duplicate titles and provide appropriate options.

5. AC5: Upload activity must be logged in the audit trail with user ID and timestamp.
Exception flow (E1): File Too Large If the uploaded file exceeds 50MB, the system displays an error: "File size exceeds limit. Maximum size: 50MB." and prevents upload. **Exception flow (E2): Database Error** If the system fails to save a document due to database error, the transaction is rolled back and an error message displayed: "System error. Please try again.

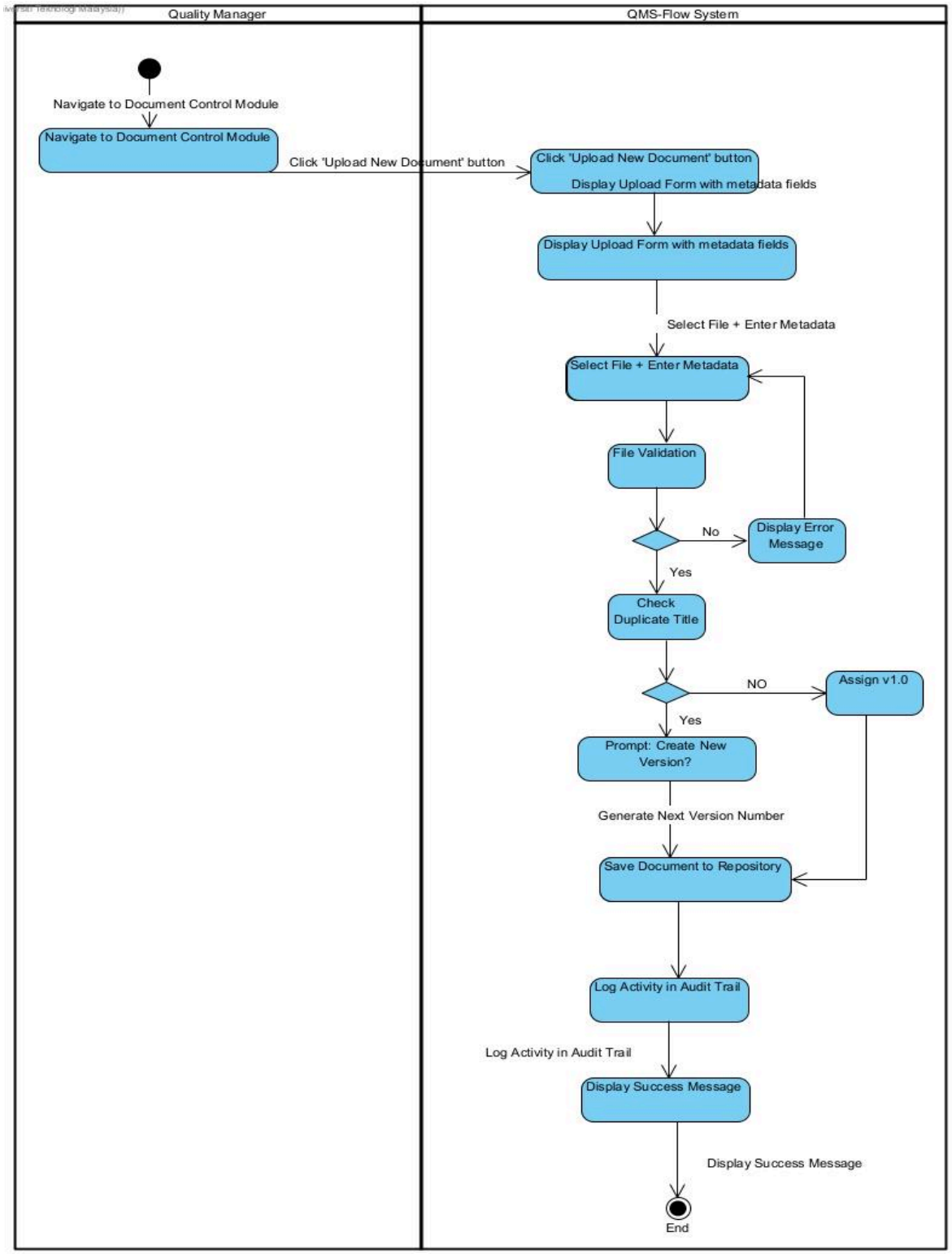

Figure 2.8: Activity Diagram for <<Manage Documents>>

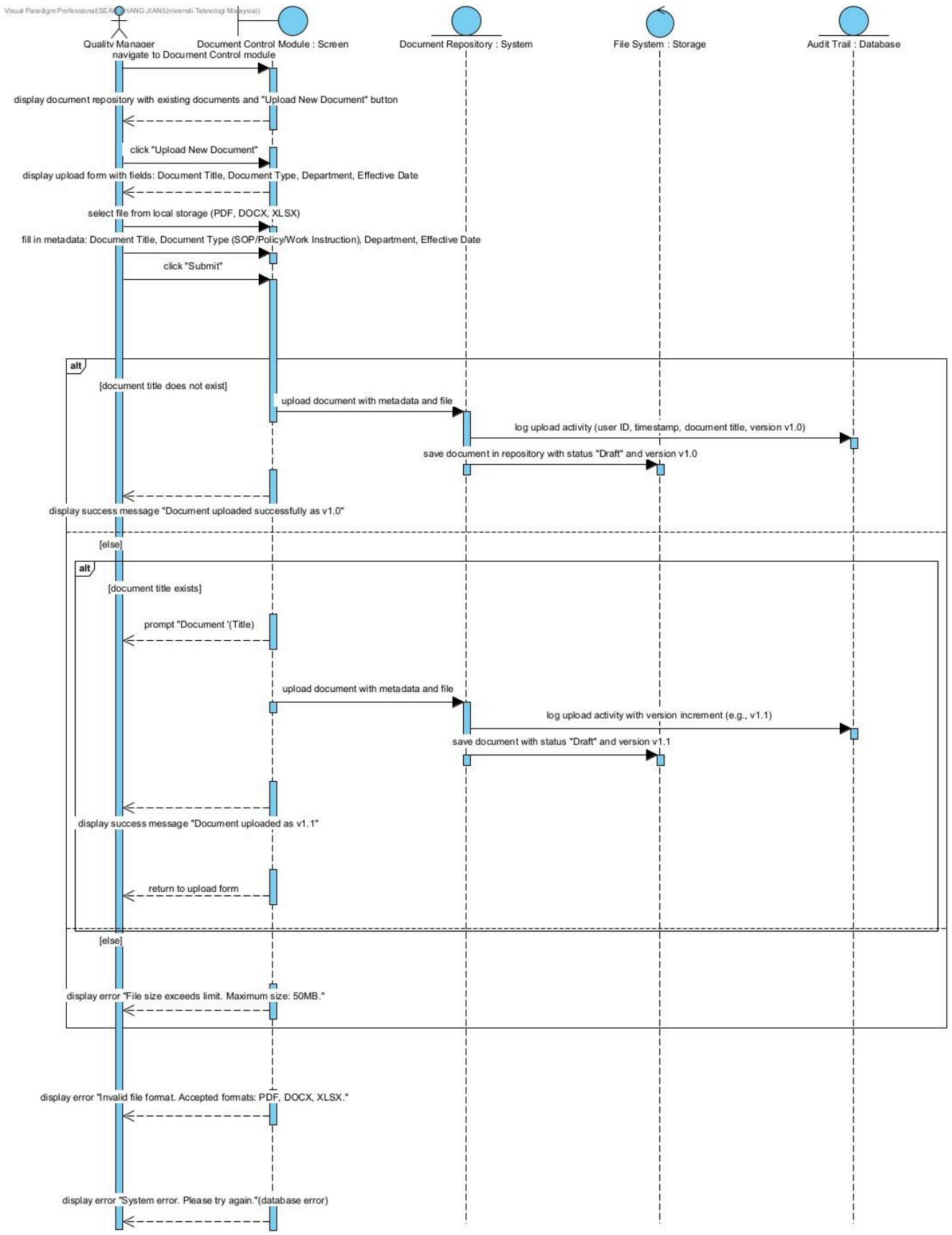

Figure 2.9: Sequence Diagram for <Manage Documents>

2.4.4 US004: User Story <Approve Changes>

Table 2.5: User Story Description for <Approve Changes>

User story: <Approve Changes> (UC04)
ID: US004
User Story Description As a Department Head, I want to review and digitally approve document changes so that documents follow proper approval workflows and are "locked" during review to prevent conflicting edits.
Preconditions: 1. Department Head is authenticated and logged in. 2. The document exists in "Under Review" status and is assigned to this Department Head. 3. The user has "Approver" role permissions for the document's department. 4. Document is locked (read-only) for all other users.
Flow of events: 1. The Department Head logs in and sees a notification badge for pending approvals. 2. Department Head clicks notification or navigates to "My Approvals" dashboard. 3. The system displays a list of documents pending approval with: Document Title, Version, Requestor, Request Date, Due Date. 4. The Department Head selects a document to review. 5. The system displays documents in a read-only viewer with change history and "Approve", "Reject", "Request Changes" buttons. 6. The Department Head reviews document content and changes the description. 7. Department Head selects action: - Approve: Clicks "Approve", optionally adds comments - Reject: Clicks "Reject", enters mandatory rejection reason - Request Changes: Clicks "Request Changes", provides feedback 8. The system validates the action and checks approval workflow requirements. 9. If "Approve" and all required approvers have approved, system: - Updates document status to "Released" - Increments version number (minor/major based on change type) - Unlocks document for general access 10. The system sends email notifications to relevant stakeholders. 11. System logs approval decision in audit trail. 12. Use case ends.
Alternative flow (A1): Partial Approval (Multi-level) If multiple approvers are required and current approval is not the final one, the system records approval, updates approval progress, and the document remains "Under Review". **Alternative flow (A2): Approval Deadline Missed** If approval is not completed by the due date, the system sends escalation notification to the backup approver or Quality Manager.
Postconditions:

1. Document status updated based on decision: Released/Rejected/Returned for Changes. 2. Version number incremented if approved and released. 3. Document unlocked/remains locked based on new status. 4. Audit trail records decision with timestamp and user. 5. Notifications sent to requestor and relevant parties.
Acceptance Criteria 1. Documents must be read-only while "Under Review". 2. System must enforce mandatory comments for Reject and Request Changes actions. 3. Version must auto-increment upon release (minor: v1.0→v1.1, major: v1.0→v2.0). 4. All stakeholders must receive email notifications upon status change. 5. Approval workflow must support multi-level approval sequences. 6. System must log complete audit trail of approval process.
Exception flow (E1): Approver Unavailable If the primary approver does not respond within 3 working days, the system escalates to backup approver automatically. **Exception flow (E2): Concurrent Approval Attempt** If two approvers try to approve the same document simultaneously, system prevents conflict with the lock mechanism.

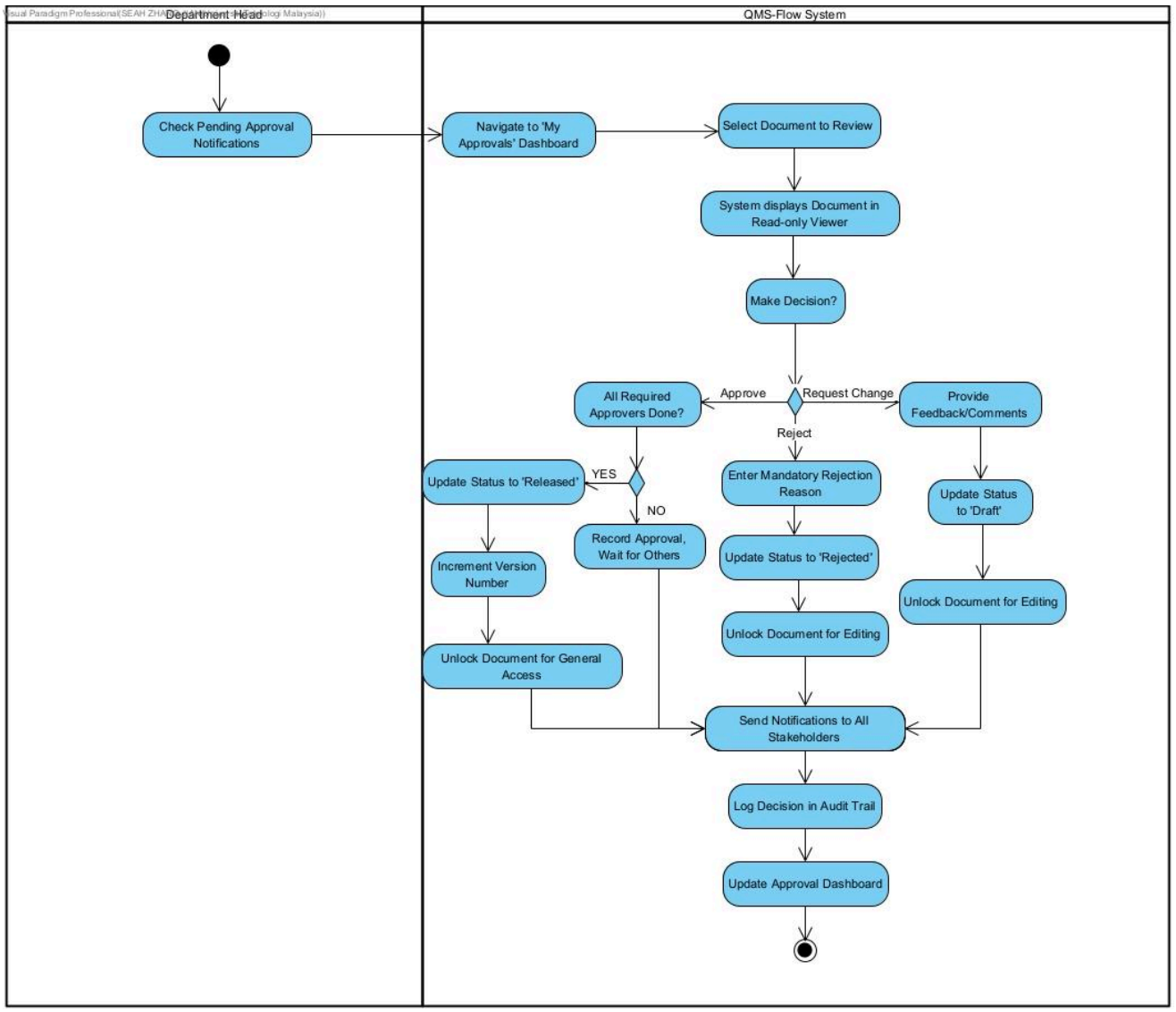

Figure 2.10: Activity Diagram for <<Approve Changes>>

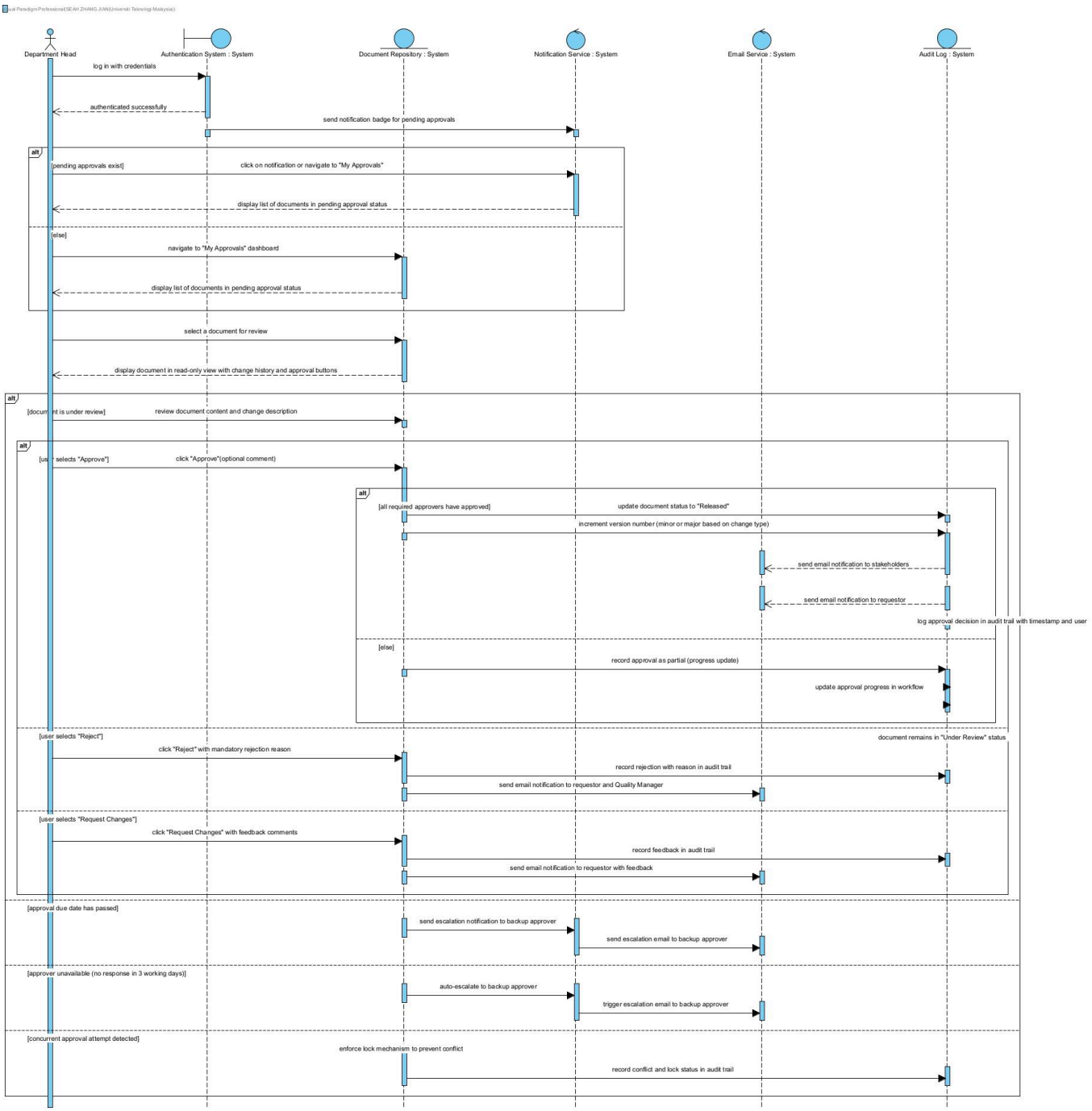

Figure 2.11: Sequence Diagram for <Approve Changes>

2.4.5 US005: User Story <Log Complaint>

Table 2.6: User Story Description for <Log Complaint>

User story: Log Complaint (UC05)
ID: US005
User Story Description As a Employee, I want to submit the complaint through a simple form so that I can report issues quickly and accurately.
Preconditions: 1. Employee is authenticated and logged in. 2. The complaint logging module is available.
Flow of events: 1. Employee navigates to Log Complaint page. 2. System displays the complaint submission form. 3. Employee enters required details: ● Customer name ● Product information ● Date of complaint ● Nature/description of issue 4. Employee optionally attaches supporting documents or photos. 5. Employee submits the complaint. 6. System validates the inputs. 7. System saves the complaint record with a unique ID. 8. Use case ends.
Alternative flow (A1*)*: Missing Required Information The system detects missing required fields and prompts the employee to complete them before resubmission.
Postconditions: 1. A new complaint record is created with status “New”. 2. Notification is sent to the Quality Manager.
Acceptance Criteria 1. Complaint Form Completeness 2. Required Field Validation 3. Attachment Upload 4. Complaint ID Generation 5. Notification to Quality Manager
Exception flow (E1): System Error During Submission If the system fails to save the complaint, an error message is shown and the employee must retry later.

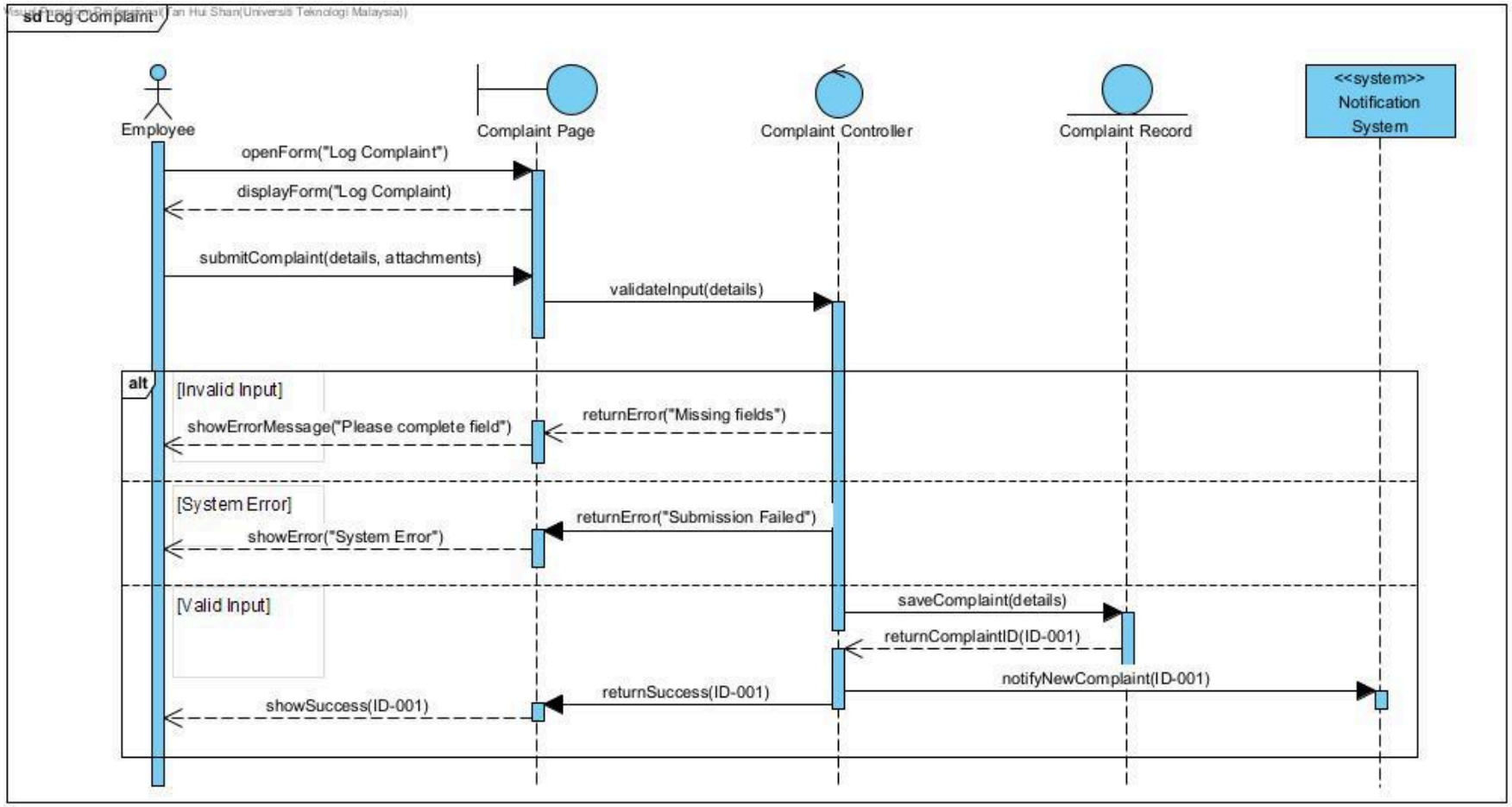

Figure 2.12: Sequence Diagram for <Log Complaint (US005)>

Figure 2.12: Sequence Diagram for <Log Complaint (US005)>

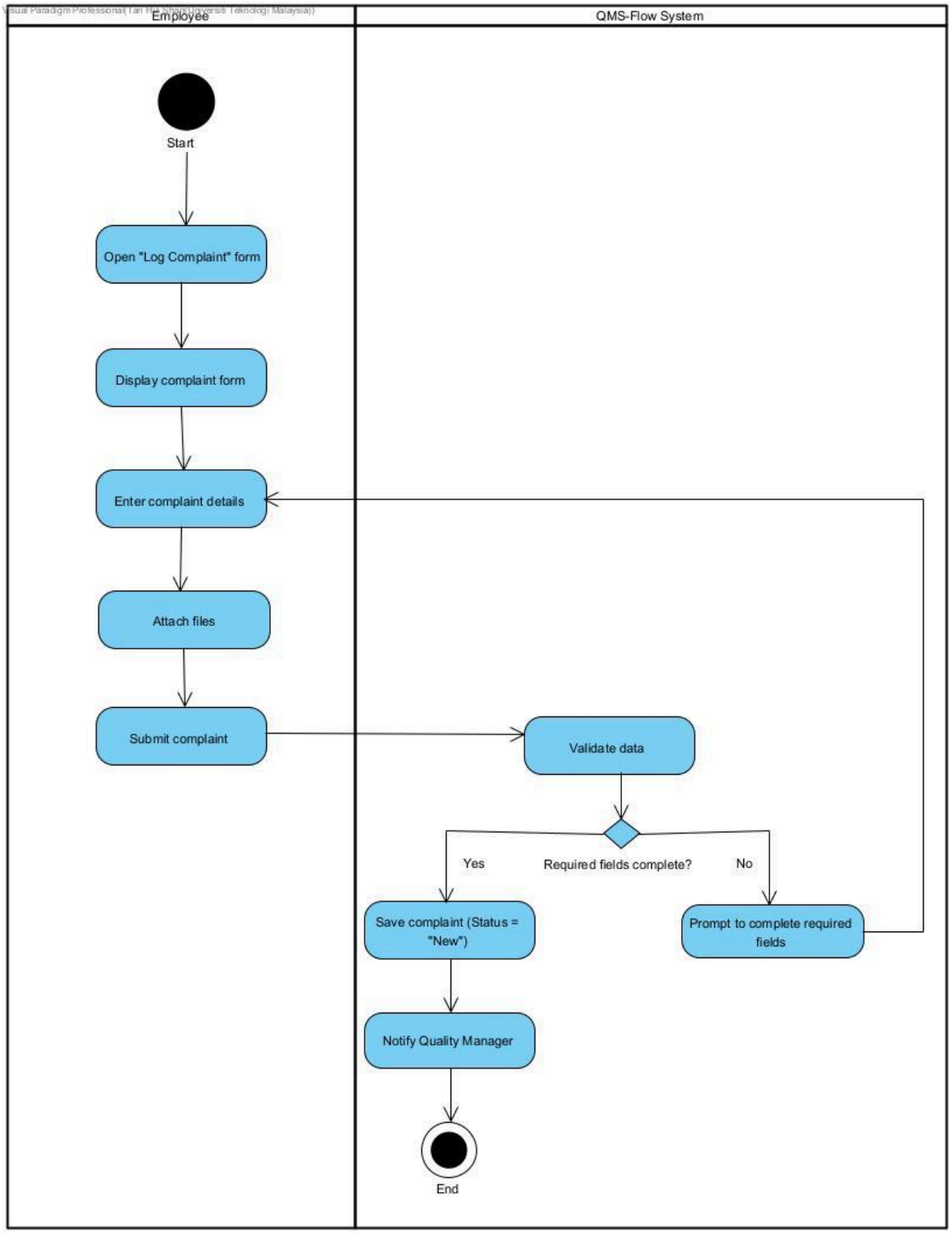

Figure 2.13: Activity Diagram for <<Log Complaint>>

2.4.6 US006: User Story <Investigate Complaint>

Table 2.7: User Story Description for <Investigate Complaint>

User story: Investigate Complaint (UC06)
ID: US006
User Story Description As a Quality Manager, I want to set a target resolution date for the assigned investigator so that the complaint can be resolved within the required timeframe.
Preconditions: 1. Quality Manager is logged in. 2. Complaint exists in “New” status.
Flow of events: 1. Quality Manager opens the Complaints Dashboard. 2. System displays complaints with status “New”. 3. Quality Manager selects a complaint to review. 4. System displays complete complaint details (customer, product, issue, attachments). 5. Quality Manager selects an investigator from the list of available staff. 6. Quality Manager enters a target resolution date. 7. Quality Manager confirms assignment. 8. Use case ends.
Alternative flow (A1): Invalid Target Date The system rejects the invalid target date and requires the Quality Manager to enter a valid one.
Postconditions: 1. Complaint is assigned to an investigator. 2. Status updated to “In Investigation”. 3. Investigator receives notification.
Acceptance Criteria: 1. Complaint Visibility 2. Complaint Assignment 3. Target Date Validation 4. Notification to Investigator 5. Access to Complaint Details 6. Prevent Duplicate Assignment
Exception flow 1: Complaint Already Under Investigation The system notifies the Quality Manager that the complaint is already assigned and prevents further reassignment.

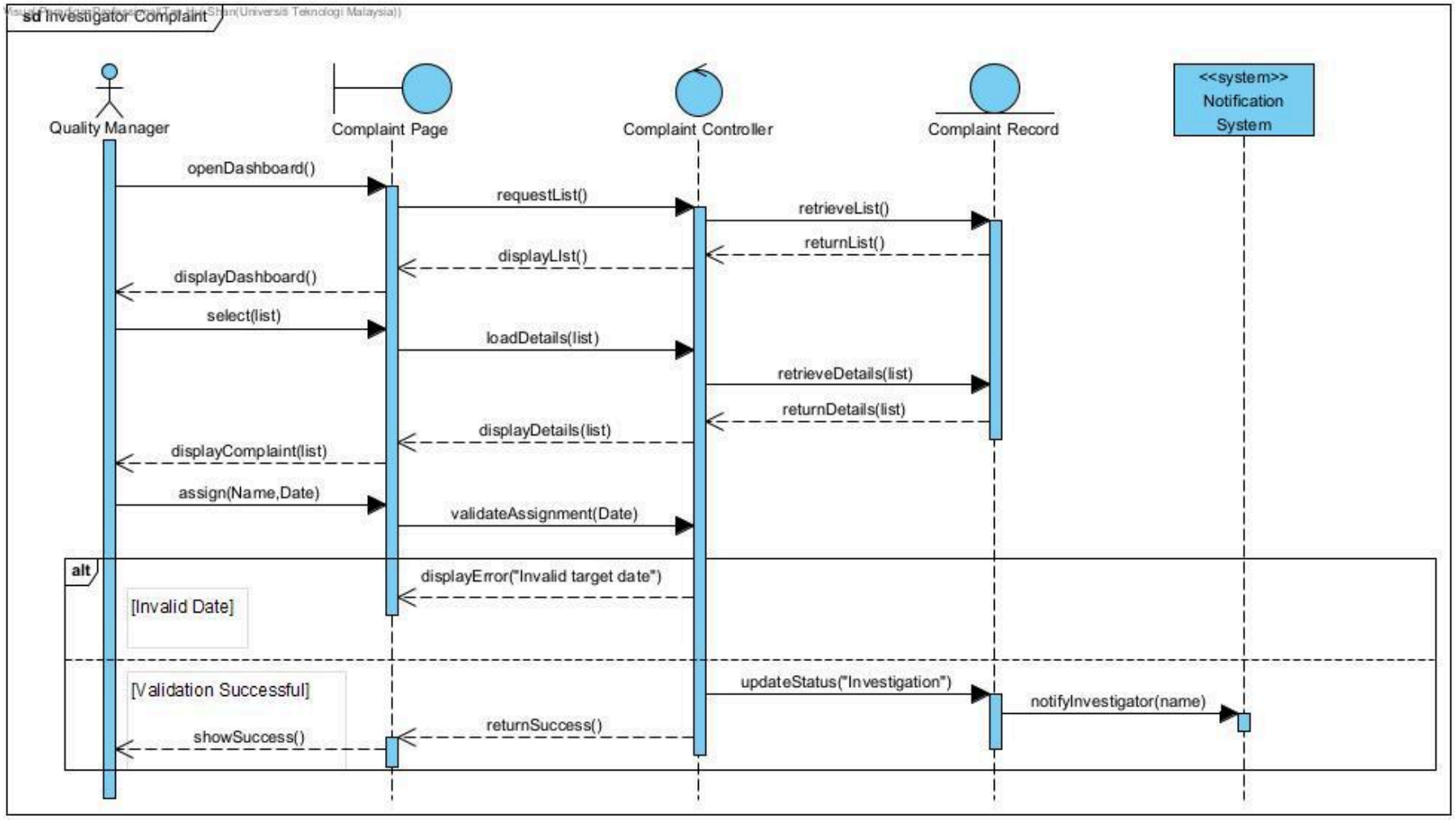

Figure 2.14: Sequence Diagram for <Investigate Complaint (US006)

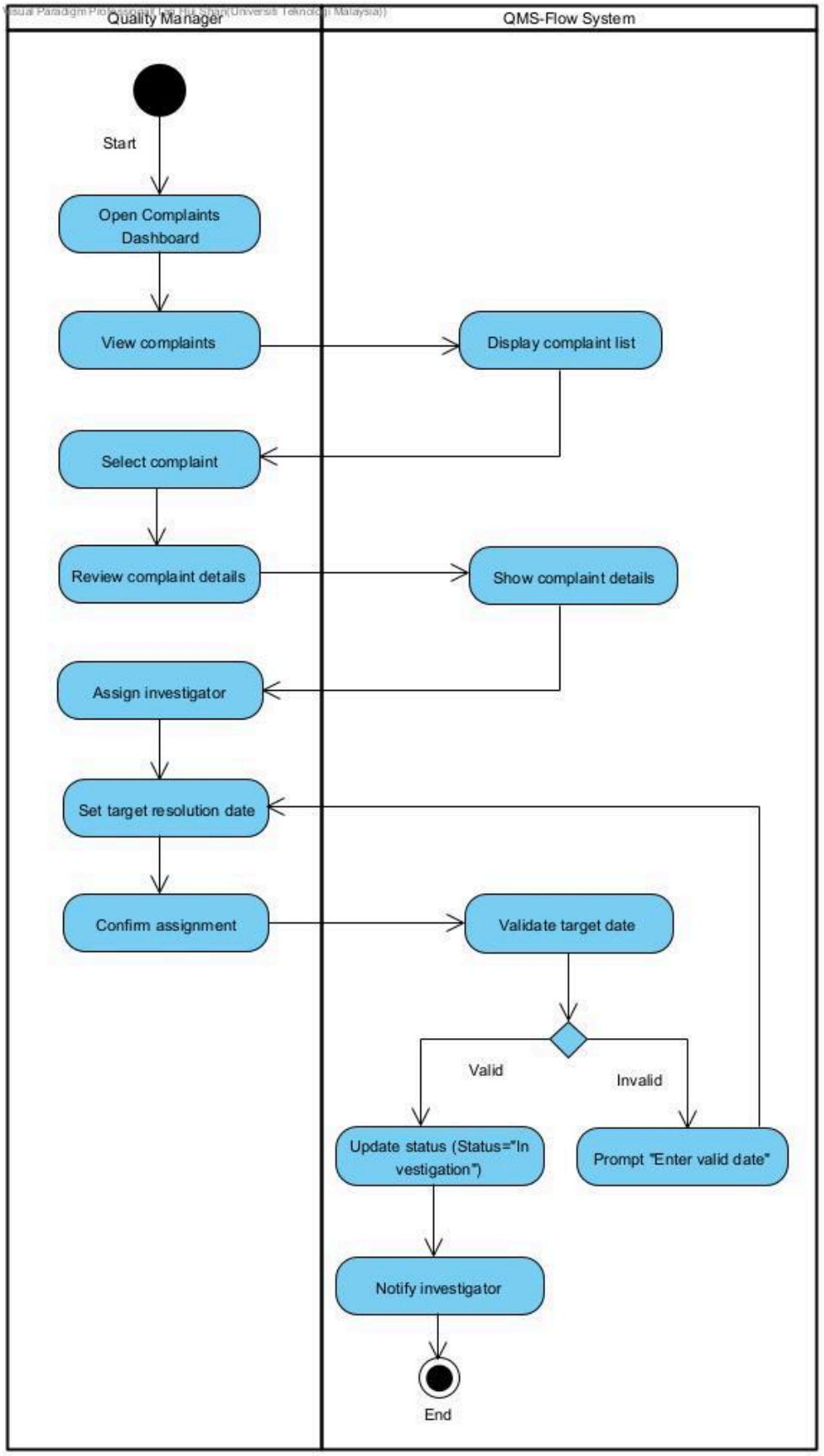

Figure 2.15: Activity Diagram for <<Investigate Complaint>>

2.4.7 US007: User Story <Initiate CAPA>

Table 2.8: User Story Description for <Initiate CAPA>

User story: Initiate CAPA (UC07)
ID: US007
User Story Description As a Quality Manager, I want to create a formal Corrective Action (CAPA) record linked to a source (Audit Finding or Customer Complaint) so that I can formally document the root cause and assign a corrective action plan.
Preconditions: 1. Quality Manager is authenticated and logged in. 2. A valid source record (Finding or Complaint) exists with a status requiring escalation.
Flow of events: 1. Quality Manager navigates to the Initiate CAPA form. 2. System prompts QM to enter the Source ID (Finding ID or Complaint ID). 3. QM enters the Root Cause Analysis (RCA) and Action Plan details. 4. QM sets the Deadline and assigns the Owner for the corrective actions. 5. QM confirms and submits the CAPA record. 6. System validates the inputs and saves the CAPA record with a unique ID. 7. System updates the status of the source record (Complaint/Finding) to 'CAPA Initiated'. 8. Use case ends.
Alternative flow (A1*)*: Invalid Source ID If the Source ID entered (Step 2) is not found, the system displays an error and requires the QM to enter a valid ID before proceeding. **Alternative Flow (A2): Missing Root Cause** The system detects that the Root Cause field is empty (Step 6) and prevents submission, prompting the QM to fill in the mandatory analysis.
Postconditions: 1. A new CAPA record is created with status "Planned". 2. The source record (Complaint/Finding) status is updated to reflect linkage to the CAPA. 3. Notification is sent to the assigned CAPA Owner (Department Head or relevant staff).
Acceptance Criteria 1. CAPA record must successfully link to a unique Finding or Complaint ID. 2. System must validate that Root Cause and Action Plan fields are completed. 3. Assigned Owner must receive an automated notification immediately upon creation.
Exception flow (E1): Database Error during CAPA Creation If the system fails to save the new CAPA record or update the source record status due to a database integrity error or connection failure, the transaction is rolled back, an error message is displayed, and the Quality Manager must retry the submission.

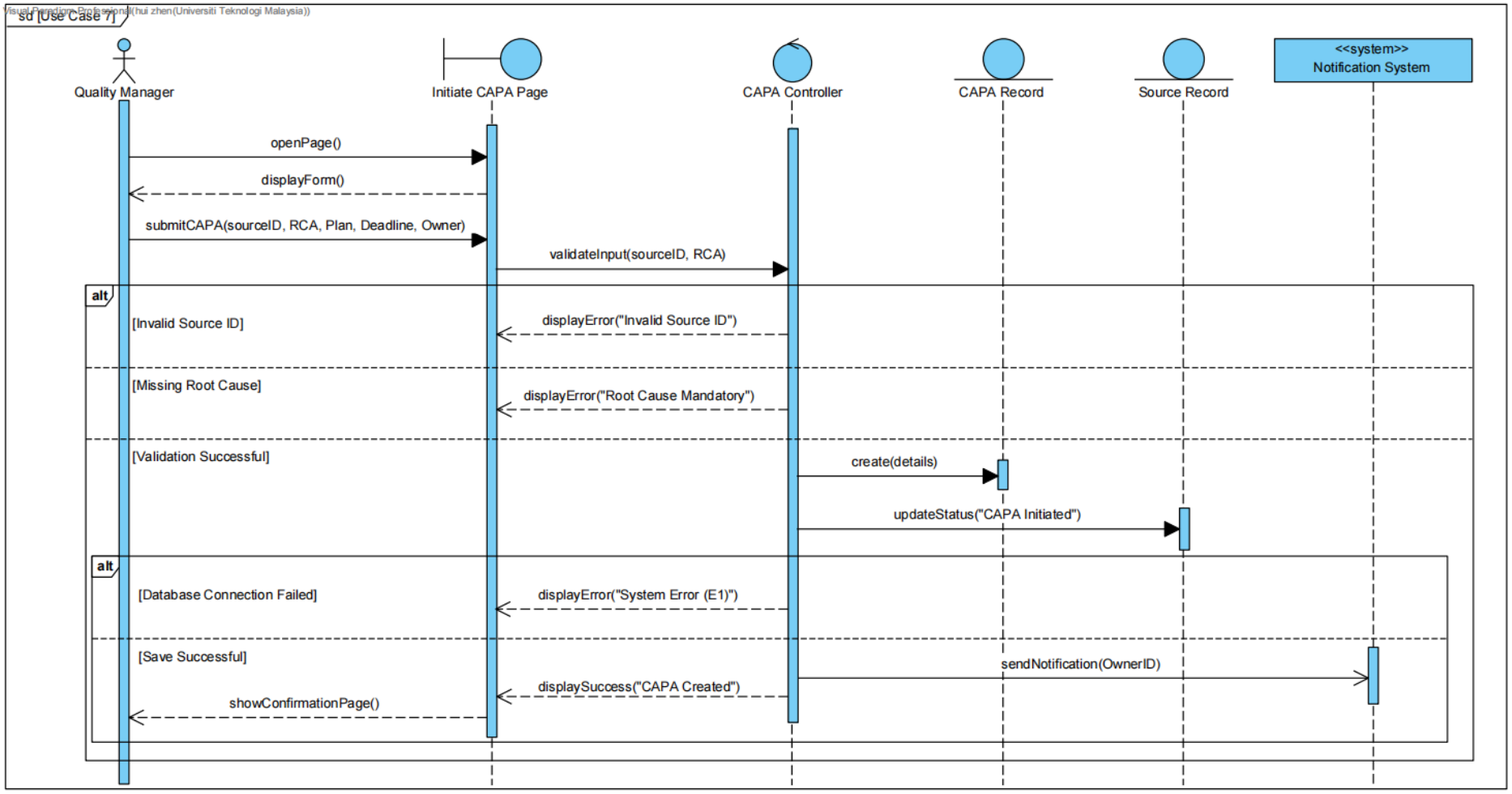

Figure 2.16: Sequence Diagram for <Initiate CAPA>

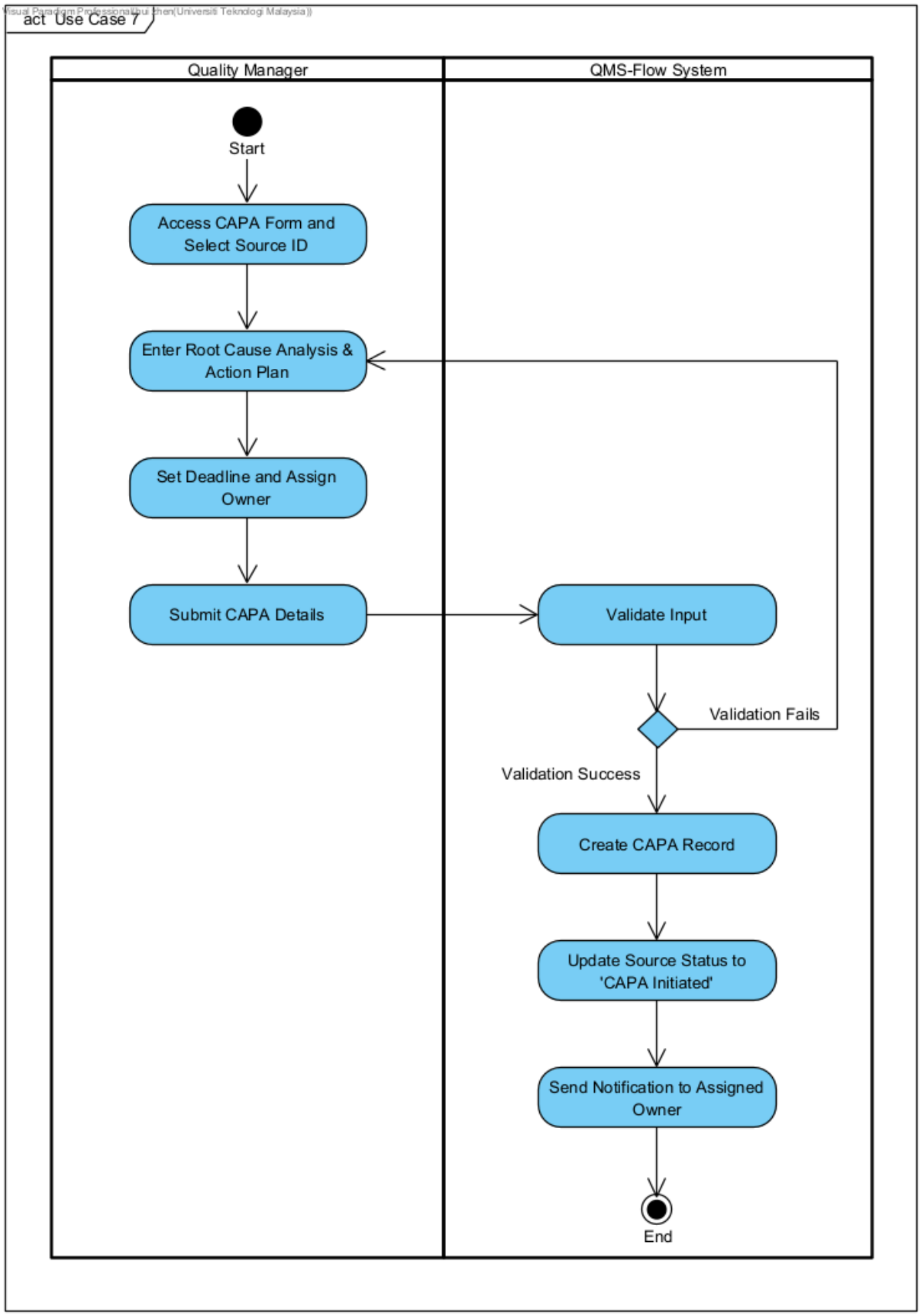

Figure 2.17: Activity Diagram for <<Initiate CAPA>>

2.4.8 US008: User Story <Verify Effectiveness>

Table 2.9: User Story Description for <Verify Effectiveness>

User story: Verify Effectiveness (UC08)
ID: US008
User Story Description As a Quality Manager, I want to record the results of the effectiveness check for a implemented CAPA so that I can confirm the root cause has been eliminated and update the record status.
Preconditions: 1. Quality Manager is logged in. 2. A CAPA record exists with status "Implemented" (Corrective actions are done). 3. The "Effectiveness Check Due Date" has arrived.
Flow of events: 1. Quality Manager navigates to the CAPA Dashboard and selects a CAPA awaiting verification. 2. System displays the CAPA details, including Root Cause and Action Plan. 3. QM reviews the evidence and enters the Verification Results and Evidence. 4. QM selects the Outcome ("Effective" or "Ineffective"). 5. QM submits the verification. 6. System validates that the mandatory evidence is attached. 7. System updates the CAPA status based on the outcome. 8. Use case ends.
Alternative flow (A1*)*: Result is Ineffective If the QM selects "Ineffective" (Step 4), the system updates the status to "Failed" and prompts the QM to either re-open the investigation or initiate a new linked CAPA. **Alternative Flow (A2): Mandatory Evidence Missing** If the QM tries to submit without entering results or attaching evidence, the system prevents submission and highlights the missing fields.
Postconditions: 1. CAPA status is updated to "Closed" (if Effective) or "Failed" (if Ineffective). 2. An automated notification is sent to the CAPA Owner regarding the final outcome.
Acceptance Criteria 1. System must restrict verification to CAPAs that are fully implemented. 2. Verification comments and evidence attachment must be mandatory. 3. Status change to "Closed" must be irreversible by standard users.
Exception flow (E1): System Failure on Status Update If the system fails to update the CAPA record status to "Closed" or "Failed" after verification (e.g., due to a server timeout), an error notification is logged internally, the Quality Manager is prompted to check the record status manually, and they must retry the action.

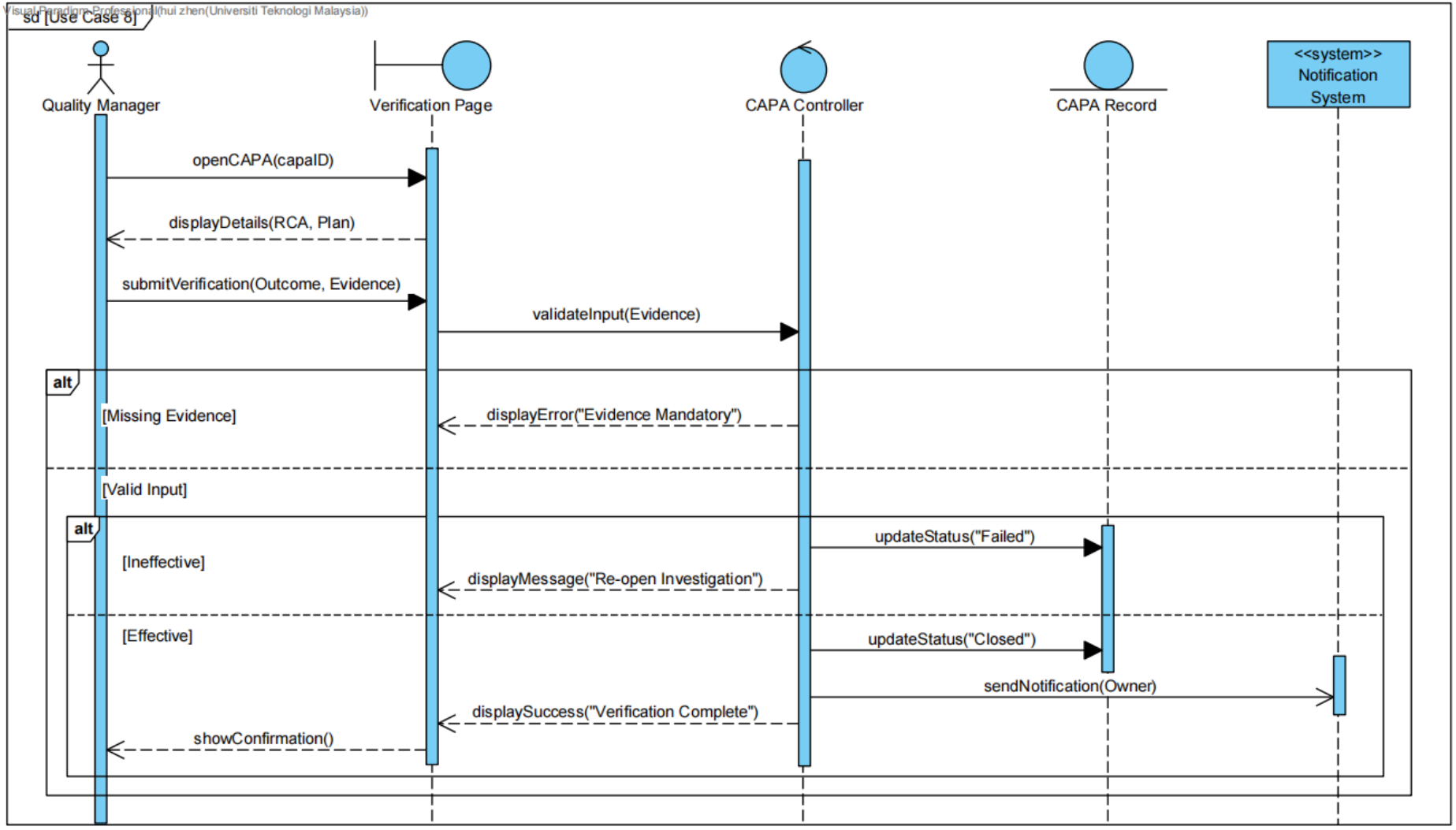

Figure 2.18: Sequence Diagram for <Verify Effectiveness>

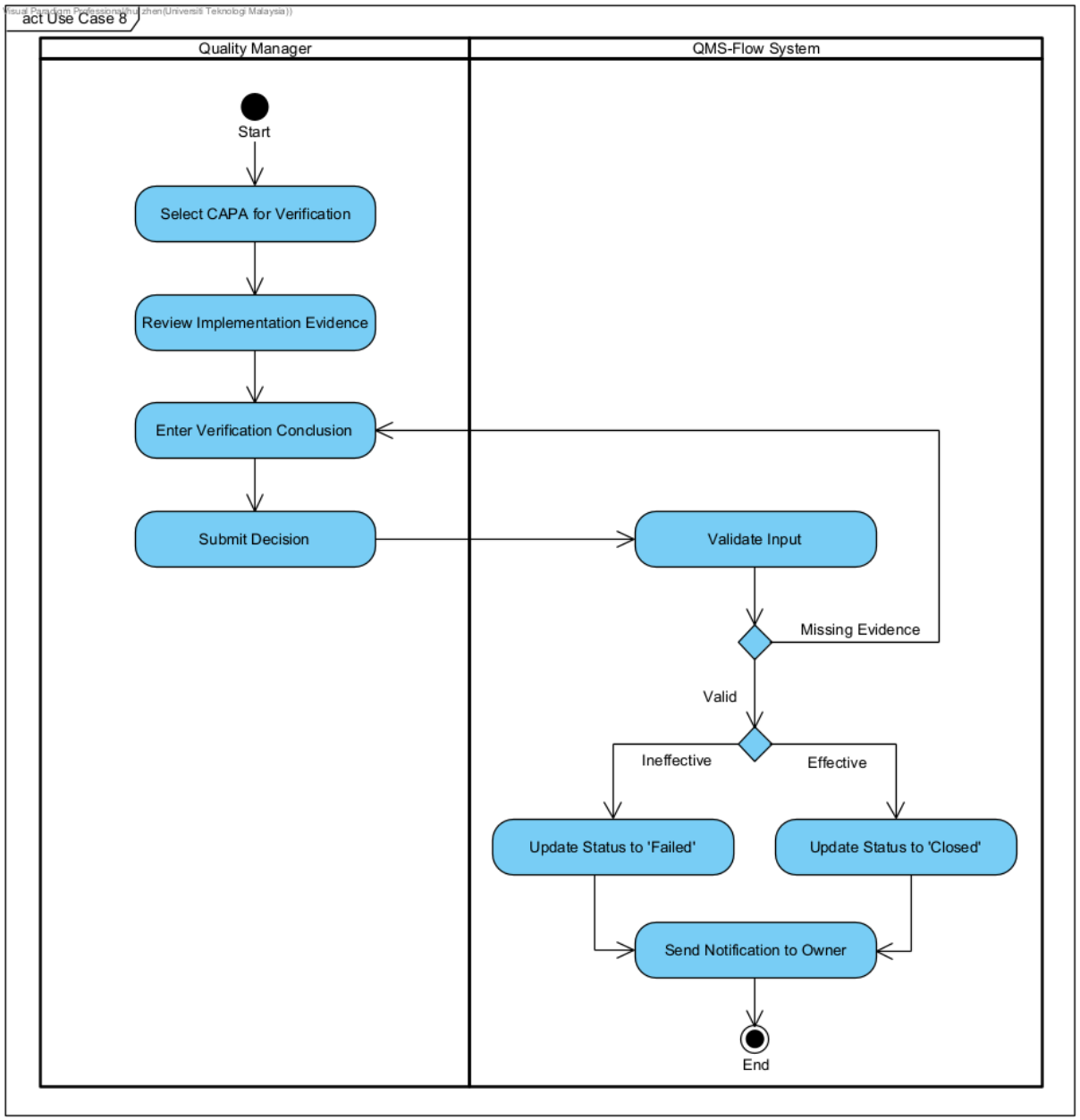

Figure 2.19: Activity Diagram for <<Verify Effectiveness>>

2.5 Performance and Other Requirements

Performance Requirement

1. **Response time:**

The system should respond to users' request and interaction such as clicks and form submission within an acceptable timeframe to ensure a smooth user experience.

2. **Throughput:**

The system should be able to support a significant amount of concurrent user at one time and also able to manage all the user requests while ensuring efficient processing and minimal delays, especially during peak periods.

3. **Capacity:**

The system should be able to store and handle a large amount of user's data such as user complaints and the approved documents without having any risk of them to be lost.

4. **Availability:**

The system should be available for 24 hours with a scheduled maintenance windows booked in advanced in order to have backups and speedy recovery plan ready for any disruptions that might occur in the future.

Software System Attributes

1. **Usability:**

The system user interface is deigned to be intuitive and user-friendly. This means it requires less than 2 hours of training for basic navigation, and it will have effective ways to handle any errors that might occur. Moreover, it shall support both English and Bahasa Malaysia.

2. **Reliability:**

The system should systematically perform well without errors or unexpected behaviours, ensuring that the content delivered is accurate and helpful to the users. It will also implement automated backups and failover mechanisms to increase the system uptime.

3. **Maintainability:**

The system should be able to be designed and implemented in a modular, documented, and well-structured manner in order to enable easy modification and improvement based on the stakeholder's needs. This will also allow new features to be deployed without a major downtime.

4. **Portability:**

The system should be able to run on different software or hardware platforms that are easily accessible by patients.

5. **Compatibility:**

The system design should be compatible with multiple kinds of devices and to provide accessibility and flexibility for the users.

<u>Other Requirements</u>

1. **Security:**

The system shall integrate with the company's existing Active Directory for user login. It will also involve encryption of data to protect all the sensitive data. Moreover, access to features and data will be strictly managed through role-based permissions.

- **Admin**: Manage system settings and user accounts
- **Quality Manager**: Oversee audits, CAPAs, and management reviews
- **Department Head**: Approve document changes and manage assigned findings
- **Employee**: Access SOPs, record training, and log complaints. Data encryption and audit trails required.

2. **Legal and Regulatory:**

All system processes will explicitly support compliance with ISO 9001:2015. A complete audit log of all users activities will be maintained for a minimum of 7 years. The system will adhere to Malaysia's Personal Data Protection Act (PDPA) 2010.

3. **Environmental:**

The system will be hosted on a major cloud platform which will be an energy-efficient data centre, minimizing its carbon footprint.

2.6 Design Constraints

Constraint Category	Description
Technical Constraints	The system must be a modern, web-based application utilizing React.js/Angular (frontend) and Node.js/Spring Boot (backend). It requires PostgreSQL 13+ with proper indexing and automated backups for data persistence. Deployment will be on Company XYZ's AWS/Azure cloud, adhering to established DevOps and security policies. The application must be cross-browser compatible and responsive for desktop and tablet use. All external integration must follow RESTful API principles, using JSON.
Performance Constraints	The system must maintain responsiveness: <3 seconds for user interaction, support 200 concurrent users without degradation, limit file uploads to 50MB, and complete compliance/analytical database queries within 10 seconds, even with 5+ years of data.
Security & Compliance Constraints	Integrate user authentication with Company XYZ's existing Active Directory/LDAP. Encrypt sensitive quality and compliance data at rest using AES-256. Strictly enforce Role-Based Access Control (RBAC) for security and data segregation. Log all system activities, including modifications and approvals, and retain for a minimum of 7 years to meet ISO 9001. Ensure full compliance with Malaysia's PDPA 2010 for handling personal data.
Document Control Constraints	Controlled documents require complete, immutable version history and detailed change tracking. Documents under formal review must be automatically locked as read-only to prevent conflicting edits. Approval workflows need configurable multi-level electronic signatures with audit trail verification. The system must generate professional PDFs and structured Excel exports suitable for external audits and regulatory compliance.
Integration Constraints	The system must be fully compatible with Company XYZ's established firewall configurations and network security policies without requiring infrastructure modifications. It must support seamless integration with Microsoft 365 services for automated email notifications, calendar scheduling, and collaborative document management. All operations must function within the existing data center environment while maintaining the contractual 99.9% uptime service level agreement (SLA) for continuous business operations.

Regulatory Constraints	All system processes and workflows must explicitly align with ISO 9001:2015 standard clauses, particularly Clauses 4 through 10 covering quality management system requirements. The system must maintain documented electronic evidence for all compliance activities with verifiable audit trails. A comprehensive multi-language interface must be provided, supporting both English and Bahasa Malaysia to accommodate all operational staff. The user interface must be fully responsive and tablet-accessible, enabling quality auditors to conduct floor inspections and document findings directly from mobile devices during facility walkthroughs.